The Bayesian Infinitesimal Jackknife for Variance: Supplemental information and proofs

A A simple closed-form example

In this section, we illustrate verification of our assumptions and the consistency of the IJ covariance with a simple closed form example, that of inferring a univariate normal mean with a misspecified variance. More elaborate examples can be found in Appendix H, including a Poisson model in the presence of overdispersed data and multilinear regression in the presence of heteroskedasticity.

For some fixed $\theta_{\mathrm{True}} \in \mathbb{R}$ and $\sigma^2 \neq 1$, let us consider the following data generating process and model:

$$\text{Data generating process:} \qquad x_n \stackrel{iid}{\sim} \mathcal{N}(\cdot|\theta_{\mathrm{True}}, \sigma^2)$$
$$\text{Model:} \qquad x_n|\theta \stackrel{iid}{\sim} \mathcal{N}(\cdot|\theta, 1) \qquad \theta \sim \mathcal{N}(\cdot|0, 1).$$

We will take θ to be our quantity of interest, so $g(\theta) = \theta$. Since $\sigma^2 \neq 1$ the model is misspecified, and so we expect the posterior variance $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}(\theta)$ to differ from V^g, even asymptotically.

For this simple model, the posterior is available in closed form:

$$\mathbb{P}(\theta|\mathcal{X}) = \mathcal{N}\left(\theta|\tilde{x}, (N+1)^{-1}\right) \quad \text{where} \quad \tilde{x} := \frac{1}{N+1}\sum_{n=1}^{N} x_n.$$

From this, we can see that $\theta_\infty = \lim_{N\to\infty} \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\theta] = \theta_{\mathrm{True}}$, It happens in this simple example that θ_∞ is "consistent" for the true mean $\underset{\mathbb{F}(x_*)}{\mathbb{E}}[x_*]$, though in general this need not be the case. For example, the model's trivial "estimate" of $\underset{\mathbb{F}(x_*)}{\mathrm{Var}}(x_*)$ is 1, which is never equal to the true value of σ^2.

One can readily verify that Assumptions 1 and 2 are satisfied for this model and this particular value of θ_∞. Arguably the most challenging condition is the final part of Assumption 1 Item 6, which one can show via strong convexity of the log likelihood, together with convergence of $\frac{1}{N}\sum_{n=1}^{N} \ell_{(1)}(x_n|\theta_\infty) \xrightarrow[N\to\infty]{prob} 0$ by the ordinary law of large numbers. Using special structure of a particular problem (like strong convexity) to establish a condition like Assumption 1 Item 6 is common in the literature — see, for example, textbook proofs of the consistency of M-estimators on unbounded domains (e.g. Theorem 9.11 of Keener (2010) or Chapter 6.2 of (Huber, 2011)).

We begin by establishing Corollary 1 and identifying our target, V^g. By the ordinary CLT applied to our closed-form expression for $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\theta]$, we have that

$$\sqrt{N}\left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\theta] - \theta_\infty\right) \xrightarrow[N\to\infty]{dist} \mathcal{N}\left(\cdot|0, \sigma^2\right),$$

from which we see that $V^g = \sigma^2$. It happens, in this case, that the variance of the limiting distribution is also the limit of the variance, since

$$N \underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}} \left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}} [\theta] \right) = \frac{N^2}{(N+1)^2} \sigma^2 \rightarrow \sigma^2 V^g.$$

The asymptotic equivalence between the variance of the limit and the limit of the variance occurs in this case because $\underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}} \left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}} [\theta] \right)$ is uniformly integrable as a sequence in N (Theorem 10.3.6 of Dudley (2018a)). We emphasize, however, that our theorems do not assume uniform integrability, and we aim to estimate the variance of the limiting normal distribution, not the limiting value of the variance, as discussed in the text surrounding Eq. 2.

To complete Corollary 1, we need to show that the expression for V^g is correct. In this case, $g_{(1)}(\theta) = 1$, and the remaining quantities are (up to constants not depending on θ):

$$\begin{aligned}
&\hat{\mathscr{L}}(x|\theta) = -\frac{1}{2N} \sum_{n=1}^{N} \left(x_n^2 - 2 x_n \theta + \theta^2 \right) + \text{Constant} \quad \text{and} \quad \log \pi(\theta) = -\frac{1}{2}\theta^2 + \text{Constant} \Rightarrow \\
&\quad \hat{\theta} = \tilde{x} \quad , \quad \ell_{(1)}(x|\theta) = x_n - \theta \quad \text{and} \quad \hat{\mathscr{L}}_{(2)}(x|\theta) = -1 \Rightarrow \\
&\quad \hat{\Sigma} = \frac{1}{N} \sum_{n=1}^{N} (x_n - \tilde{x})^2 \quad , \quad \hat{\mathcal{I}} = 1 \quad \text{and} \quad \hat{V}^g = \hat{\mathcal{I}}^{-1} \hat{\Sigma} \hat{\mathcal{I}}^{-1} = \hat{\Sigma}. \qquad (1)
\end{aligned}$$

It is readily verified that $\hat{\Sigma} \xrightarrow[N \rightarrow \infty]{prob} \sigma^2 = V^g$, so we have verified the conclusion of Corollary 1.

Finally, we manually verify Theorem 2. The influence score is given by:

$$\begin{aligned}
\psi_n &= N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\theta, -\frac{1}{2} \left(x_n^2 - 2 x_n \theta + \theta^2 \right) \right) \\
&= -\frac{1}{2} N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} (\theta, 1)\, x_n^2 + N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} (\theta, \theta)\, x_n + N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\theta, \theta^2 \right) = \frac{N}{N+1} x_n + \text{Constant},
\end{aligned}$$

where the constant in the final line is the same for all x_n. Since the IJ covariance depends only on the difference $\psi_n - \bar{\psi}$, quantities that are the same for each n can be neglected in the IJ covariance computation. Therefore the IJ covariance estimate is given by

$$V^{\mathrm{IJ}} = \frac{1}{N} \sum_{n=1}^{N} \left(\psi_n - \bar{\psi} \right)^2 = \frac{N^2}{(N+1)^2} \hat{\Sigma} \xrightarrow[N \rightarrow \infty]{prob} \sigma^2 = V^g.$$

Thus V^{IJ} differs from the $\hat{V}^g$ given in Eq. 1 by a factor that vanishes asymptotically. So, like $\hat{V}^g$, V^{IJ} is a consistent estimator of the asymptotic variance V^g.

B Notation key

For ease of reference, we provide here a table of key quantities defined and used through the course of our proofs.

Symbol	Description	Definition
ψ_n	Influence score of the n–th observation	Eq. 3
$\mathcal{S}$	A set of sets of data indices	Definition 1
$\mathcal{X}_s$	The observations corresponding to $s \in \mathcal{S}$	Definition 1
$\phi(\theta, \mathcal{X}_s)$	A function of interest depending on θ and a set of datapoints	Definition 2
B_δ	A δ-ball around θ_∞	Assumption 1
$\mathscr{T}_{(k)}\left(\phi, \hat{\theta}, \theta\right)$	The k–th order Taylor series residual term for ϕ	Definition 3
$\hat{\mathcal{G}}(\tau)$	Gaussian density with covariance $\hat{\mathcal{I}}^{-1}$	Definition 8
$\mathcal{G}_\lambda^S(\tau)$	Isotropic Gaussian density with precision λ truncated to set S	Definition 8
$\hat{Z}$	Normalizing constant for $\hat{\mathcal{G}}(\tau)$	Definition 8
$\lambda_{\mathcal{I}}, \hat{\lambda}_{\mathcal{I}}$	Minimum eigenvalues of $\mathcal{I}$, $\hat{\mathcal{I}}$	Definition 4
τ	Centered and scaled parameter $\sqrt{N}(\theta - \hat{\theta})$	Eq. 10
R_1, R_2, R_3	Inner, middle, and outer regions of integration	Definition 7
δ_1	Scale of boundary between R_1 and R_2	Definition 7
δ_2	Scale of boundary between R_2 and R_3	Definition 7
$I(\phi)$	The integration functional of ϕ	Eq. 14
$I_k(\phi)$	Integration functional restricted to region R_k	Eq. 15
$\mathfrak{R}$	The event that the problem is "regular"	Lemma 4
ϵ_U	Arbitrarily small accuracy parameter for the regular event	Lemma 4
ϵ_L	Log likelihood lower bound in regular event	Lemma 4
$\mathscr{R}^\phi(\mathcal{X}_s)$	The BCLT residual for $\underset{\mathbb{P}(\theta\mid\mathcal{X})}{\mathbb{E}}[\phi(\theta, \mathcal{X}_s)]$	Definition 9
$\mathscr{R}^k(\mathcal{X}_s)$	The k–th component of the BCLT residual	Definition 9

Table 1: Notation key

C Setup and Preliminaries

C.0.1 Derivative arrays

Throughout our proofs, we will be forming and manipulating higher-order Taylor expansions of functions of θ. The arrays of derivatives in these expressions will always be summed against a direction, typically $\tau = \sqrt{N}(\theta - \hat{\theta})$. In order to avoid cumbersome summation notation for all such terms, we will implicitly sum derivative arrays against τ or θ. or example, for a function $\psi(\theta, x_n) \in \mathbb{R}^{D_\psi}$, we will write

$$\psi_{(3)}(\hat{\theta}, x_n)\tau^3 = \sum_{a=1}^{D}\sum_{b=1}^{D}\sum_{c=1}^{D} \left.\frac{\partial^3 \psi(\theta, x_n)}{\partial\theta_a \partial\theta_b \partial\theta_c}\right|_{\hat{\theta}} \tau_a \tau_b \tau_c.$$

Occasionally we will need to sum several derivative arrays against τ, particularly in Section D.7. Since the directions τ will always be the same, we will combine the τ in such expressions with no ambiguity. For example, for the term involving $\hat{\mathscr{L}}_{(3)}(\hat{\theta})$ in the statement of Theorem 1, we will write

$$\psi_{(1)}(\hat{\theta},x)\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^4 = \sum_{a=1}^{D}\sum_{b=1}^{D}\sum_{c=1}^{D}\sum_{d=1}^{D}\psi_{(1:a)}(\hat{\theta},x)\hat{\mathscr{L}}_{(3:b,c,d)}(\hat{\theta})\tau_a\tau_b\tau_c\tau_d.$$

When necessary, higher derivatives will be indexed in the parentheses. For example, $\psi_{(2:i,j)}(\theta, x_n)$ will denote the length–D_ψ vector

$$\left.\frac{\partial^2\psi(\theta,x_n)}{\partial\theta_i\partial\theta_j}\right|_{\theta,x_n}.$$

We denote by $\mathbb{I}(A)$ the indicator function taking value 1 when the event A occurs and 0 otherwise.

We will use order notation to describe the behavior of quantities as $N \to \infty$. For example, we will use $z = O(\phi(N))$ to mean there exists an N^* and an $A > 0$ such that $N > N^* \Rightarrow z < A\phi(N)$. In our notation, A and N^* in this definition will never depend on the data, $\mathcal{X}$, though they may depend on unobserved population quantities. Dependence on the data $\mathcal{X}$ in order notation will always be represented with O_p notation. For example, $z = O_p(\phi(N))$ will mean that there exists an $A > 0$ such that for any $\epsilon > 0$, there is a corresponding N^* and such that

$$N > N^* \quad \Rightarrow \quad \mathop{\mathbb{E}}_{\mathbb{F}(\mathcal{X})}\left[\mathbb{I}\left(z < A\phi(N)\right)\right] > 1-\epsilon.$$

Here, A does not depend on ϵ. The meaning is that the event $z = O(\phi(N))$ occurs with arbitrarily high probability as N goes to infinity. The notation $\tilde{O}$ and $\tilde{O}_p$ will have the same meaning as O and O_p notation respectively, but with factors of $\log N$ possibly ignored.

C.0.2 General index sets

To succinctly state our theorems for double sums, it will be convenient to introduce a more general index notation.

Definition 1. Let $\mathcal{S}$ denote a set of sets of data indices, where each member $s \in \mathcal{S}$ has the same cardinality, and for $s \in \mathcal{S}$, let $\mathcal{X}_s$ denote the set of corresponding datapoints. We write $\phi(\mathcal{X}_s)$ for a function that depends on $|s|$ datapoints. For example, if $\mathcal{S} = \{1,\ldots,N\} =: [N]$, then $\mathcal{X}_s = x_s$, and $\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\phi(\mathcal{X}_s) = \frac{1}{N}\sum_{n=1}^{N}\phi(x_n)$. Similarly, if $\mathcal{S} = [N]\times[N]$, then $\mathcal{X}_s = (x_{s_1}, x_{s_2})$, and $\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\phi(\mathcal{X}_s) = \frac{1}{N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}\phi(x_n,x_m)$. ◀

We now state a generalization of Definition 2 that applies to generic index sets as defined in Definition 1.

Definition 2. For an index set $\mathcal{S}$, a data-dependent function $\phi(\theta, \mathcal{X}_s)$ is K-th order BCLT-okay if, for each $k = 1,\ldots,K$,

1. $\theta \mapsto \phi_{(k)}(\theta,\mathcal{X}_s)$ is BCLT-okay $\mathbb{F}$-almost surely
2. $\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\mathop{\mathbb{E}}_{\pi(\theta)}\left[\|\phi(\theta,\mathcal{X}_s)\|_2^2\right] = O_p(1)$.
3. For some $\delta > 0$, $\sup_{\theta\in B_\delta}\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\left\|\phi_{(k)}(\theta,\mathcal{X}_s)\right\|_2^2 = O_p(1)$.

◀

Note that a function is BCLT–okay according to Definition 2 precisely if it is BCLT–okay according to Definition 2 with index set $\mathcal{S} = [N]$. Although $\mathcal{S} = [N]$ and $\mathcal{S} = [N] \times [N]$ will be the only examples of $\mathcal{S}$ we will use in the present paper, we believe it is clearer to keep the notation general.

In particular, Theorem 1 is stated in terms of $\mathcal{S} = [N]$, but we will prove that it holds for generic index sets of the form Definition 2. Formally, we will prove Theorem 1 under the following Assumption 1, noting that Assumption 1 is assumed by Theorem 1 for $\mathcal{S} = [N]$.

Assumption 1. *Assume that $\phi(\theta, \mathcal{X}_s)$ is third-order BCLT-okay according to Definition 2, and that Assumption 1 holds.*

C.1 Taylor series expansions and their asymptotics

The proofs below repeatedly rely on two key ideas: Taylor series expansions, and uniform asymptotic control on the terms that arise. In this section, we introduce these ideas and prove two key lemmas. The first, Lemma 1, gives the form of a Taylor series expansion. The second, Lemma 2, controls the asymptotic behavior of the log likelihood and its derivatives in a neighborhood of θ_∞.

C.1.1 Taylor series expansions

Our analysis is based strongly on Taylor series expansions. We will use the following two minor variants of standard analysis tools. The first is a multivariate version of the integral form of the Taylor series residual for a vector-valued function.

Definition 3. For a k-times differentiable function $\phi(\theta)$, a fixed $\hat{\theta}$, and and θ, define the residual functional

$$\mathscr{T}_{(k)}\left(\phi, \hat{\theta}, \theta\right) := \frac{1}{(k-1)!}\int_0^1 (1-t)^{k-1}\phi_{(k)}(\hat{\theta} + t(\theta - \hat{\theta}))dt.$$

◀

The residual $\mathscr{T}_{(k)}\left(\phi, \hat{\theta}, \theta\right)$ is an array of the same dimension as the k-th order derivative, $\phi_{(k)}(\theta)$. Observe that, for any norm $\|\cdot\|$,

$$\left\|\mathscr{T}_{(k)}\left(\phi, \hat{\theta}, \theta\right)\right\| \le \frac{1}{(k-1)!} \sup_{\tilde{\theta}:\left\|\tilde{\theta}-\hat{\theta}\right\|_\infty \le \left\|\theta-\hat{\theta}\right\|_\infty} \left\|\phi_{(k)}(\tilde{\theta})\right\|.$$

Lemma 1. *A K-times continuously differentiable function $\phi(\theta)$ has Taylor series residual*

$$\phi(\theta) = \sum_{k=0}^{K-1} \frac{1}{k!}\phi_{(k)}(\hat{\theta})(\theta - \hat{\theta})^k + \mathscr{T}_{(K)}\left(\phi, \hat{\theta}, \theta\right)(\theta - \hat{\theta})^K.$$

Proof. Let $\phi_d(\theta)$ denote the d-th component of the vector $\phi(\theta)$ (contrast this notation with for the d-th order derivative $\phi_{(d)}(\theta)$). Consider the scalar to scalar map $t \mapsto \phi_d(\hat{\theta} + t(\theta - \hat{\theta}))$. Taking the integral remainder form of the $K-1$-th order Taylor series expansion of this scalar map Dudley, 2018b, Appendix B.2 and stacking the result into a vector gives the desired result. □

C.1.2 Log likelihood asymptotics

To use Lemma 1 to study $\hat{\mathscr{L}}(\theta)$, we will need to expand Definition 1 with the following quantities.

Definition 4.

$$\hat{\mathscr{L}}_{(k)}(\theta) := \frac{1}{N}\sum_{n=1}^{N} \ell_{(k)}(x_n|\theta) + \frac{1}{N}\left(\log \pi\right)_{(k)}(\theta) \quad \mathscr{L}_{(k)}(\theta) := \int \ell_{(k)}(x_*|\theta) d\mathbb{F}(x_*)$$

$$\lambda_{\mathcal{I}} := \text{The minimum eigenvalue of } \mathcal{I} \qquad \hat{\lambda}_{\mathcal{I}} := \text{The minimum eigenvalue of } \hat{\mathcal{I}}$$

Note that, following our convention, $\hat{\mathscr{L}}_{(k)}(\theta)$ is the k–th partial derivative of $\hat{\mathscr{L}}(\theta)$ with respect to θ.

The term $\mathscr{L}_{(k)}(\theta)$ is a potential abuse of notation, since it is only the partial derivative of $\mathscr{L}(\theta)$ under the assumption that we can interchange integration and differentiation. Though we will in fact be assuming sufficient conditions for this interchange (Assumption 1 Item 5), our proofs will simply use the above definition of $\mathscr{L}_{(k)}(\theta)$ directly. ◀

Using Lemma 1, we can then form a Taylor series expansion around $\hat{\theta}$, an expression that we will analyze carefully in Appendix D:

$$\begin{aligned}\hat{\mathscr{L}}(\theta) = \hat{\mathscr{L}}(\hat{\theta}) + \hat{\mathscr{L}}_{(1)}(\hat{\theta})(\theta - \hat{\theta}) + \frac{1}{2}\hat{\mathscr{L}}_{(2)}(\hat{\theta})(\theta - \hat{\theta})^2 + \\ \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})(\theta - \hat{\theta})^3 + \mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)(\theta - \hat{\theta})^4.\end{aligned} \tag{2}$$

We now turn to the task of controlling the kinds of terms that arise in Eq. 2. One of our key tools in the proof of Lemma 2 will be *uniform laws of large numbers* (ULLNs), a concept which we make concrete in Definition 5. Typically, ULLNs require that the sample average $\frac{1}{N}\sum_{n=1}^{N}\phi(x_n, \theta)$ converge to $\underset{\mathbb{F}(x_*)}{\mathbb{E}}\left[\phi(x_n, \theta)\right]$ uniformly over θ in some compact set.

Definition 5. Recall the definition of B_δ from Assumption 1. We say a "uniform law of large numbers", or "ULLN", holds for a function $\phi(\theta, x_*)$ if, for some $\delta > 0$,

$$\sup_{\theta \in B_\delta} \left\| \frac{1}{N}\sum_{n=1}^{N} \phi(\theta, x_n) - \underset{\mathbb{F}(x_*)}{\mathbb{E}}\left[\phi(\theta, x_*)\right] \right\|_2 \xrightarrow[N\to\infty]{prob} 0,$$

where $\underset{\mathbb{F}(x_*)}{\mathbb{E}}\left[\phi(\theta, x_*)\right]$ is finite. ◀

In general, a ULLN requires that the set of functions $\{\theta \mapsto \phi(\theta, \mathcal{X}) : \|\theta - \theta_\infty\|_2 < \delta\}$ be a "Glivenko-Cantelli class" with finite covering number (Chapter 19 of van der Vaart (2000), van der Vaart and Wellner (2013)). A simple condition for a uniform law of large numbers to hold is that $\phi(\theta, x_*)$ be dominated by an absolutely integrable function of x_* alone in a compact neighborhood of θ_∞ (Example 19.7 of van der Vaart (2000)).

We now state a key lemma controlling the asymptotic behavior of the log likelihood and its derivatives.

Lemma 2. *Suppose that there exists $\delta_{LLN} > 0$, $M(x_*)$, and $K \geq 1$ such that*

$$\sup_{\theta \in B_{\delta_{LLN}}} \left\| \ell_{(K)}(x_* | \theta) \right\|_2 \leq M(x_*) \quad \textit{with} \quad \mathop{\mathbb{E}}_{\mathbb{F}(x_*)} \left[M(x_*)^2 \right] \leq \infty. \tag{3}$$

Addionally assume that $\pi(\theta)$ is non-zero and $K-1$ times continuously differentiable in $B_{\delta_{LLN}}$. Then the following results hold. First, for $k \in \{0, \ldots, K\}$,

$$\sup_{\theta \in B_{\delta_{LLN}}} \frac{1}{N} \sum_{n=1}^{N} \left\| \ell_{(k)}(x_n | \theta) \right\|_2^2 = O_p(1). \tag{4}$$

Then, for $k \in \{0, \ldots, K-1\}$,

$$\sup_{\theta \in B_{\delta_{LLN}}} \left\| \frac{1}{N} \sum_{n=1}^{N} \left(\ell_{(k)}(x_n | \theta) - \mathop{\mathbb{E}}_{\mathbb{F}(x_*)} \left[\ell_{(k)}(x_* | \theta) \right] \right) \right\|_2 \xrightarrow[N \to \infty]{prob} 0 \tag{5}$$

$$\sup_{\theta \in B_{\delta_{LLN}}} \left\| \frac{1}{\sqrt{N}} \sum_{n=1}^{N} \left(\ell_{(k)}(x_n | \theta) - \mathop{\mathbb{E}}_{\mathbb{F}(x_*)} \left[\ell_{(k)}(x_* | \theta) \right] \right) \right\|_2 = O_p(1) \tag{6}$$

$$\sup_{\theta \in B_{\delta_{LLN}}} \left\| \hat{\mathscr{L}}_{(k)}(\theta) - \mathscr{L}_{(k)}(\theta) \right\|_2 \xrightarrow[N \to \infty]{prob} 0. \tag{7}$$

Proof. First, note that

$$\begin{aligned}
&\sup_{\theta \in B_{\delta_{LLN}}} \frac{1}{N} \sum_{n=1}^{N} \left\| \ell_{(K)}(x_n | \theta) \right\|_2^2 \leq \frac{1}{N} \sum_{n=1}^{N} \sup_{\theta \in B_{\delta_{LLN}}} \left\| \ell_{(K)}(x_n | \theta) \right\|_2^2 \quad \text{and} \\
&\mathop{\mathbb{E}}_{\mathbb{F}(x_*)} \left[\sup_{\theta \in B_{\delta_{LLN}}} \left\| \ell_{(K)}(x_* | \theta) \right\|_2^2 \right] \leq \mathop{\mathbb{E}}_{\mathbb{F}(x_*)} \left[M(x_*)^2 \right] < \infty,
\end{aligned}$$

so Eq. 4 holds for $k = K$ by the strong law of large numbers.

We now prove the remaining relations inductively. For a scalar–valued, differentiable function $f(\theta)$, and any two θ, θ' in $B_{\delta_{LLN}}$, we can write

$$\begin{aligned}
|f(\theta) - f(\theta')| &= \left| \int_0^1 \left. \frac{\partial f(\theta' + t(\theta - \theta'))}{\partial t} \right|_t dt \right| \\
&= \left| \int_0^1 f_{(1)}(\theta' + t(\theta - \theta'))(\theta - \theta') dt \right| \\
&\leq \|\theta - \theta'\|_2 \sup_{\tilde{\theta} \in B_{\delta_{LLN}}} \left\| f_{(1)}(\tilde{\theta}) \right\|_2.
\end{aligned}$$

Applying the previous formula componentwise to $\ell_{(K-1)}(x_* | \theta)$, and loosening the right hand side to contain all of $\left\|\ell_{(K)}(x_*|\tilde{\theta})\right\|_2$ rather than only a particular component, gives

$$\begin{aligned}\left\|\ell_{(K-1)}(x_*|\theta)\right\|_\infty &\le \|\theta-\theta'\|_2 \sup_{\tilde{\theta}\in B_{\delta_{LLN}}} \left\|\ell_{(K)}(x_*|\tilde{\theta})\right\|_2 \\ &\le \|\theta-\theta'\|_2 M(x_*). \end{aligned} \tag{8}$$

It follows that each component of $\ell_{(K-1)}(x_*|\theta)$ is both Glivenko-Cantelli and Donsker, and so Eqs. 5 and 6 hold for $k = K-1$ (see van der Vaart, 2000, Example 19.7, as well as the discussion in Definition 5 above). Further,

$$\sup_{\theta\in B_{\delta_{LLN}}} \left\|\ell_{(K-1)}(x_*|\theta)\right\|_2 \le \sup_{\theta\in B_{\delta_{LLN}}} \sqrt{D_\theta}\left\|\ell_{(K-1)}(x_*|\theta)\right\|_\infty \le 2\sqrt{D_\theta}\delta_{LLN} M(x_*),$$

where the final line applies Eq. 8 and $\sup_{\theta,\theta'\in B_{\delta_{LLN}}} \|\theta-\theta'\|_2 \le 2\delta_{LLN}$. It follows that Eq. 4 holds for $k = K-1$.

Since Eq. 3 holds for $k = K-1$, the remaining results follow by proceeding inductively on $K-1$, $K-2$, and so on.

Finally, by continuity of $\pi(\theta)$ (Assumption 1 Item 2) and compactness of $B_{\delta_{LLN}}$, we have that, as $N\to\infty$,

$$\frac{1}{N}\sup_{\theta\in B_{\delta_{LLN}}} \left\|\left(\log\pi\right)_{(k)}(\theta)\right\|_2^2 \to 0.$$

Then Eq. 7 follows from the preceding display, Cauchy-Schwarz, and Eq. 5. □

C.2 Consistency results

Finally, we prove known results on the frequentist behavior of key quantities using our own assumptions and notation.

Lemma 3. *Under Assumption 1,* $\hat{\theta} \xrightarrow[N\to\infty]{prob} \theta_\infty$*, and* $\left\|\hat{\mathcal{I}}^{-1}\right\|_{op} < 2\lambda_{\mathcal{I}}^{-1}$ *, with probability approaching one.*

Proof. First, by strict optimality of the log likelihood (Assumption 1 Item 6) and the boundedness of the prior density (Assumption 1 Item 2), we must have $\hat{\theta}\in B_\delta$ with probability approaching one as $N\to\infty$, since

$$\begin{aligned}
&\sup_{\theta\in\Omega_\theta\setminus B_\delta} \hat{\mathscr{L}}(\theta_\infty) - \hat{\mathscr{L}}(\theta) \\
&\quad= \sup_{\theta\in\Omega_\theta\setminus B_\delta} \frac{1}{N}\sum_{n=1}^N \ell(x_n|\theta_\infty) - \frac{1}{N}\sum_{n=1}^N \ell(x_n|\theta) + \frac{1}{N}\log\pi(\theta_\infty) - \frac{1}{N}\log\pi(\theta) \\
&\quad\ge \sup_{\theta\in\Omega_\theta\setminus B_\delta} \frac{1}{N}\sum_{n=1}^N \ell(x_n|\theta_\infty) - \frac{1}{N}\sum_{n=1}^N \ell(x_n|\theta) - \frac{2}{N}\sup_{\theta\in\Omega_\theta}|\log\pi(\theta)| \\
&\quad\ge \varepsilon + \frac{2}{N}\sup_{\theta\in\Omega_\theta}|\log\pi(\theta)| && \text{(Assumption 1 Item 6)} \\
&\quad\ge \varepsilon + O\left(N^{-1}\right). && \text{(Assumption 1 Item 3)}
\end{aligned}$$

By definition of $\hat{\theta}$, $\hat{\mathscr{L}}(\theta_\infty) - \hat{\mathscr{L}}(\hat{\theta}) \leq 0$, and so $\hat{\theta} \notin \Omega_\theta \setminus B_\delta$ for sufficiently large N. It therefore suffices to consider the behavior of $\hat{\theta} \in B_\delta$.

By Assumption 1 Item 6, $\mathcal{I}$ is positive definite with $\|\mathcal{I}^{-1}\|_{op} = \left\|\mathscr{L}_{(2)}(\theta_\infty)\right\|_{op} = \lambda_{\mathcal{I}}^{-1}$. Choose δ small enough that $\mathscr{L}_{(2)}(\theta)$ is continuous by Assumption 1 Item 4, then choose δ smaller still so that, by continuity of the operator norm,

$$\sup_{\theta \in B_\delta} \left\|\mathscr{L}_{(2)}(\theta)^{-1}\right\|_{op} < \sqrt{2}\lambda_{\mathcal{I}}^{-1}.$$

Next, choose δ sufficiently small that, by the ULLN of Assumption 1 Item 5 and Lemma 2,

$$\sup_{\theta \in B_\delta} \left\|\hat{\mathscr{L}}_{(2)}(\theta)^{-1}\right\|_{op} < 2\lambda_{\mathcal{I}}^{-1},$$

with probability approaching one. For the remainder of the proof, we will condition on the event that the preceding display holds.

Using Lemma 1 to Taylor expand $\hat{\mathscr{L}}_{(1)}$ around $\hat{\theta}$ and evaluate at θ_∞ gives

$$\hat{\mathscr{L}}_{(1)}(\theta_\infty) = \hat{\mathscr{L}}_{(1)}(\hat{\theta}) + \mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right)(\hat{\theta} - \theta_\infty). \tag{9}$$

Combining the above results, the minimum eigenvalue of the matrix $\mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right)$ is controlled by

$$\begin{aligned}
\inf_{v:\|v\|_2=1} v^T \mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right) v = & \inf_{v:\|v\|_2=1} \int_0^1 (1-t) v^T \hat{\mathscr{L}}_{(2)}(\hat{\theta} + t(\theta_\infty - \hat{\theta})) v dt \\
\geq & \inf_{v:\|v\|_2=1} \inf_{\theta \in B_\delta} v^T \hat{\mathscr{L}}_{(2)}(\theta) v \\
\geq & \frac{\lambda_{\mathcal{I}}}{2}.
\end{aligned}$$

Consequently, the matrix $\mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right)$ is invertible with $\left\|\mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right)^{-1}\right\|_{op} \leq 2\lambda_{\mathcal{I}}^{-1}$. Applying to the Taylor expansion Eq. 9, and using the fact that $\hat{\mathscr{L}}_{(1)}(\hat{\theta}) = 0$, gives that

$$\left\|\hat{\theta} - \theta_\infty\right\|_2 \leq 4\lambda_{\mathcal{I}}^{-1} \left\|\hat{\mathscr{L}}_{(1)}(\theta_\infty)\right\|_2.$$

Finally, by a LLN applied to $\hat{\mathscr{L}}_{(1)}$, we have $\hat{\mathscr{L}}_{(1)}(\theta_\infty) \xrightarrow[N\to\infty]{prob} \mathscr{L}_{(1)}(\theta_\infty) = 0$, concluding the proof. □

Proof of Proposition 1.

Proof. The proof uses Eq. 9 in the proof of Lemma 3. First, let $\delta_N := \left\|\hat{\theta} - \theta_\infty\right\|_2$, so that, by Lemma 3, $\delta_N \xrightarrow[N\to\infty]{prob} 0$.

Then, observe that

$$
\begin{aligned}
&\left\| \mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right) - \mathscr{L}_{(2)}(\theta_\infty) \right\|_2 \\
&\quad = \left\| \int_0^1 (t-1)\left(\hat{\mathscr{L}}_{(2)}(t(\theta_\infty - \hat{\theta}) + \hat{\theta})dt - \mathscr{L}_{(2)}(\theta_\infty)\right) dt \right\|_2 \\
&\quad \le \sup_{\theta \in B_{\delta_N}} \left\| \hat{\mathscr{L}}_{(2)}(\theta) - \mathscr{L}_{(2)}(\theta_\infty) \right\|_2 \\
&\quad \le \sup_{\theta \in B_{\delta_N}} \left\| \hat{\mathscr{L}}_{(2)}(\theta) - \hat{\mathscr{L}}_{(2)}(\theta_\infty) \right\|_2 + \sup_{\theta \in B_{\delta_N}} \left\| \hat{\mathscr{L}}_{(2)}(\theta_\infty) - \mathscr{L}_{(2)}(\theta_\infty) \right\|_2 \\
&\quad \xrightarrow[N\to\infty]{prob} 0,
\end{aligned}
$$

where the limit to zero follows from continuity of $\hat{\mathscr{L}}_{(2)}$ via Assumption 1 Item 4 for the first term and the ULLN via Assumption 1 Item 5 and Lemma 2 for the second term. It follows that

$$
\mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right)^{-1} \xrightarrow[N\to\infty]{prob} \mathscr{L}_{(2)}(\theta_\infty)^{-1} = -\mathcal{I}^{-1}.
$$

Next, by Assumption 1 Item 5 and Lemma 2, $\underset{\mathbb{F}(x)}{\mathbb{E}}\left[\left\|\ell_{(1)}(x|\theta_\infty)\right\|_2^2\right]$ is finite with probability approaching one, so $\underset{\mathbb{F}(x)}{\mathrm{Cov}}\left(\ell_{(1)}(x|\theta_\infty)\right) = \Sigma$ is finite. By smoothness of $\mathscr{L}$ (Assumption 1 Item 4) and the definition of θ_∞, $\underset{\mathbb{F}(x)}{\mathbb{E}}\left[\ell_{(1)}(x|\theta_\infty)\right] = 0$. The x_n are IID, so, by a central limit theorem,

$$
\sqrt{N}\hat{\mathscr{L}}_{(1)}(\theta_\infty) = \frac{1}{\sqrt{N}} \sum_{n=1}^{N} \ell_{(1)}(x_n|\theta_\infty) \xrightarrow[N\to\infty]{dist} \mathcal{N}(\cdot|0, \Sigma).
$$

By Eq. 9 in the proof of Lemma 3, with probability approaching one,

$$
\begin{aligned}
\sqrt{N}(\hat{\theta} - \theta_\infty) &= \mathscr{T}_{(2)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta_\infty\right)^{-1} \sqrt{N}\hat{\mathscr{L}}_{(1)}(\theta_\infty) \\
&\xrightarrow[N\to\infty]{dist} \mathcal{N}(\cdot|0, \mathcal{I}^{-1}\Sigma\mathcal{I}^{-1}),
\end{aligned}
$$

where the final line follows from Slutsky's theorem. □

D Proof of Theorem 1

The proof of Theorem 1 will be broken up into several parts. Our first step is to derive an explicit series expansion for posterior expectations of the form $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\phi(\theta)]$, culminating in Lemma 11. In Lemma 14, we then consider the dependence of $\phi(\theta)$ on a subset of datapoints, writing $\phi(\theta, \mathcal{X}_s)$ (recalling Definition 1), and showing that the results of Lemma 11 can be summed over $\mathcal{S}$. The residual of the expansion in Lemma 11 is designed so that Lemma 14 follows easily.

To prove Lemma 11, we follow classical proofs of the BCLT, keeping explicit track of the residual. We first express the desired expectation as the ratio of integrals with respect to τ. We then divide the domain of integration into three regions according to their distance from $\hat{\theta}$ and treat each separately. Finally, we combine the bounds on the integrals over different regions for our final result.

Throughout, we will use the change of variable

$$\theta := \hat{\theta} + \sqrt{N}\tau \quad \Leftrightarrow \quad \tau = N^{-1/2}(\theta - \hat{\theta}). \tag{10}$$

We will switch the domain of integration between τ and θ without re-writing the arguments. For example, for some function $\phi(\theta)$, we will write the shorthand $\int \phi(\theta) d\tau$ for the more precise $\int \phi\left(\sqrt{N}\tau + \hat{\theta}\right) d\tau$. Similarly, we will overload set notation: if A is a set of θ, we will write $\tau \in A$ as a shorthand for $\tau \in \left\{\tau : \hat{\theta} + \sqrt{N}\tau \in A\right\}$. Although slight abuses of notation, these conventions will keep our expressions more compact, as well as serve as a reminder of the θ domain in which the integrand is smooth by assumption.

D.1 Ratio of integrals

We need to consider the asymptotic behavior of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\phi(\theta, \mathcal{X}_s)]$, which can be written as the ratio of the integrals

$$\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\phi(\theta, \mathcal{X}_s)] = \frac{\int \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta)\right) d\theta}{\int \exp\left(N\hat{\mathscr{L}}(\theta)\right) d\theta}.$$

Recall that $\hat{\theta} := \mathrm{argmax}_\theta \hat{\mathscr{L}}(\theta)$ and define $\tilde{\theta}(t) := t(\theta - \hat{\theta}) + \hat{\theta}$. We will employ the following Taylor series expansion:

$$\begin{aligned}\hat{\mathscr{L}}(\theta) = \hat{\mathscr{L}}(\hat{\theta}) + \hat{\mathscr{L}}_{(1)}(\hat{\theta})(\theta - \hat{\theta}) + \frac{1}{2}\hat{\mathscr{L}}_{(2)}(\hat{\theta})(\theta - \hat{\theta})^2 + \\ \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})(\theta - \hat{\theta})^3 + \mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)(\theta - \hat{\theta})^4,\end{aligned} \tag{11}$$

recalling the definition of the integral form of the Taylor series residual $\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)$ given in Definition 3.

Note that $\hat{\mathscr{L}}(\hat{\theta})$ is constant in θ, and that $\hat{\mathscr{L}}_{(1)}(\hat{\theta}) = 0$. Plugging in and changing the domain of integration to τ, we see

$$\begin{aligned}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\phi(\theta, \mathcal{X}_s)] &= \frac{\int \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\theta}{\int \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\theta} \\ &= \frac{\int \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) \sqrt{N} d\tau}{\int \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) \sqrt{N} d\tau}.\end{aligned}$$

Since the factor of $\sqrt{N}$ cancels in the preceding display, it will suffice to consider the following generic integral.

Definition 6. For a target function ϕ, define the integral functional

$$I(\phi) := \int \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau \tag{12}$$

$$= \int \phi(\theta, \mathcal{X}_s) \times \tag{13}$$

$$\exp\left(\frac{1}{2}\hat{\mathscr{L}}_{(2)}(\hat{\theta})\tau^2 + N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^3 + N^{-1}\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)\tau^4\right) d\tau. \tag{14}$$

◀

Under Definition 6, $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\phi(\theta, \mathcal{X}_s)\right] = I(\phi)/I(1)$. We will now consider the behavior of $I(\phi)$ for a generic third-order BCLT-okay ϕ, returning to consider the behavior of the ratio in Section D.8.

D.2 Regions

For some δ_1 and δ_2 to be determined later, we will divide consideration of $I(\phi)$ into the following three disjoint regions.

Definition 7. For any $\delta_1 > 0$ and $\delta_2 > 0$ and N sufficiently large, define the disjoint regions

$$\begin{aligned}
R_1 &:= \left\{\theta : \left\|\theta - \hat{\theta}\right\|_2 < \delta_1 \frac{\log N}{\sqrt{N}}\right\} \\
R_2 &:= \left\{\theta : \delta_1 \frac{\log N}{\sqrt{N}} \le \left\|\theta - \hat{\theta}\right\|_2 < \delta_2\right\} \\
R_3 &:= \left\{\theta : \delta_2 \le \left\|\theta - \hat{\theta}\right\|_2\right\}.
\end{aligned}$$

Recalling our slight abuse of set notation, the same regions in the domain of τ are,

$$\begin{aligned}
\tau \in R_1 &\Rightarrow \tau \in \left\{\tau : \|\tau\|_2 < \delta_1 \log N\right\} \\
\tau \in R_2 &\Rightarrow \left\{\tau : \delta_1 \log N \le \|\tau\|_2 < \delta_2\sqrt{N}\right\} \\
\tau \in R_3 &\Rightarrow \left\{\tau : \delta_2\sqrt{N} \le \|\tau\|_2\right\}.
\end{aligned}$$

Similarly, we will use the regions as domains of integration for τ as well as θ. Correspondingly, we can define

$$\begin{aligned}
I_1(\phi) &:= \int_{R_1} \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau \\
I_2(\phi) &:= \int_{R_2} \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau \\
I_3(\phi) &:= \int_{R_3} \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau.
\end{aligned} \tag{15}$$

◀

Since the boundaries of the regions coincide, $I(\phi) = I_1(\phi) + I_2(\phi) + I_3(\phi)$ for N when the regions R_1, R_2, and R_3 are disjoint, which will occur for N sufficiently large (see Lemma 4 below for more discussion of this point). Note that the union $R_1 \bigcup R_2$ is a δ_2 ball around $\hat{\theta}$, and that R_1 shrinks to $\hat{\theta}$ as $N \to \infty$.

Throughout the proof, we will be choosing δ_1 and δ_2 to satisfy certain conditions. Our assumptions will always take the form of assuming that δ_1 is *large enough*, and that δ_2 is *small enough*. For any particular δ_1 and δ_2, we then will require that N is large enough that the regions are disjoint.

D.3 Nearly certain events

Our entire analysis will be conditioned on the occurrence of several events, each of which occurs with probability approaching one as N goes to infinity. Outside of these events we provide no guarantees. As a consequence, the residuals of our approximation can be used to show convergence in probability or distribution (via Slutsky's theorem), but not convergence in expectation nor almost sure convergence.

Lemma 4. *For any $\epsilon_U > 0$, let $\mathfrak{R}$ denote the event that all of the following occur for a particular N, δ_1, δ_2. ("R" is for "regular".)*

1. *The regions R_1, R_2, and R_3 are disjoint, and $I(\phi) = I_1(\phi) + I_2(\phi) + I_3(\phi)$.*
2. $\left\Vert \hat{\theta} - \theta_\infty \right\Vert_\infty < \epsilon_U$.
3. $|\pi(\hat{\theta}) - \pi(\theta_\infty)| < \epsilon_U$.
4. *For any scalar-valued function $\rho(\theta, x_n)$ with $\underset{\mathbb{F}(x_*)}{\mathbb{E}}[\rho(\theta, x_*)] = 0$ satisfying a ULLN with $\delta = \delta_{LLN}$, we have $\sup_{\theta \in R_1 \bigcup R_2} \frac{1}{N} \sum_{n=1}^N |\rho(\theta, x_n)| \le \epsilon_U$.*
5. $\hat{\lambda}_\mathcal{I} > \lambda_\mathcal{I}/2$.
6. *There exists $\epsilon_L > 0$ such that $\sup_{\theta \in R_3} \left(\sum_{n=1}^N \ell(x_n|\hat{\theta}) - \sum_{n=1}^N \ell(x_n|\theta) \right) > N\epsilon_L$.*

Let Assumption 1 hold. For any $\delta_1 > 0$, any $\pi_0 \in (0, 1)$, and any $\delta_2 > 0$ and $\epsilon_U > 0$ satisfying $\delta_{LLN} > \epsilon_U + \delta_2$ and $\delta_2/2 \ge \epsilon_U$, there exists an N^* such $N > N^*$ implies that event $\mathfrak{R}$ occurs with $\mathbb{F}$-probability at least $1 - \pi_0$.

Proof. Event 1 follows from the fact that, for any positive δ_1 and δ_2, eventually $\delta_2 \sqrt{N} > \delta_1 \log N$.

Event 2 follows from Lemma 3.

Event 3 follows from the fact that the prior density is continuous by Assumption 1 Item 2 and, by Lemma 3, we can take $\hat{\theta}$ to be arbitrarily close to θ_∞.

For Event 4, observe that $\left\|\hat{\theta} - \theta_\infty\right\|_2 < \epsilon_U$ implies that

$$
\begin{aligned}
\theta \in & R_1 \bigcup R_2 \Rightarrow \\
\left\|\theta - \hat{\theta}\right\|_2 \le & \delta_2 \Rightarrow \\
\|\theta - \theta_\infty\|_2 \le & \left\|\theta - \hat{\theta}\right\|_2 + \left\|\hat{\theta} - \theta_\infty\right\|_2 \le \delta_2 + \epsilon_U \le \delta_{LLN}.
\end{aligned}
$$

Therefore,

$$
\sup_{\theta \in R_1 \bigcup R_2} \frac{1}{N} \sum_{n=1}^N |\rho(\theta, x_n)| \le \sup_{\theta \in B_{\delta_{LLN}}} \frac{1}{N} \sum_{n=1}^N |\rho(\theta, x_n)| \xrightarrow[N \to \infty]{prob} 0.
$$

Event 5 follows from Lemma 3.

For Event 6, write

$$
\begin{aligned}
\frac{1}{N} \sup_{\theta \in R_3} \left(\sum_{n=1}^N \ell(x_n|\hat{\theta}) - \sum_{n=1}^N \ell(x_n|\theta) \right) = & \\
\hat{\mathscr{L}}(\hat{\theta}) - \log \pi(\hat{\theta}) - \sup_{\theta \in R_3} \left(\hat{\mathscr{L}}(\theta) - \log \pi(\theta) \right) = & \\
\hat{\mathscr{L}}(\hat{\theta}) - \hat{\mathscr{L}}(\theta_\infty) + \sup_{\theta \in R_3} \left(\hat{\mathscr{L}}(\theta_\infty) - \hat{\mathscr{L}}(\theta) \right). &
\end{aligned}
$$

Since $\left\|\hat{\theta} - \theta_\infty\right\|_2 \le \epsilon_U \le \delta_2/2$ by assumption of the present lemma,

$$
\begin{aligned}
\theta \in & R_3 \Rightarrow \\
\left\|\theta - \hat{\theta} + \hat{\theta} - \theta_\infty\right\|_2 \ge & \delta_2 \Rightarrow \\
\left\|\theta - \hat{\theta}\right\|_2 + \left\|\hat{\theta} - \theta_\infty\right\|_2 \ge & \delta_2 \Rightarrow \\
\left\|\theta - \hat{\theta}\right\|_2 \ge & \delta_2 - \frac{\delta_2}{2} = \frac{\delta_2}{2},
\end{aligned}
$$

so the set R_3, is contained in $\Omega_\theta \setminus B_{\delta_2/2}$. Consequently, by Assumption 1 Item 6, there exists an ϵ_L such that

$$
\sup_{\theta: \left\|\theta - \hat{\theta}\right\|_2 \ge \delta_2} \left(\hat{\mathscr{L}}(\theta_\infty) - \hat{\mathscr{L}}(\theta) \right) \le \sup_{\theta \in \Omega_\theta \setminus B_{\delta_2/2}} \left(\hat{\mathscr{L}}(\theta_\infty) - \hat{\mathscr{L}}(\theta) \right) \le -2\epsilon_L.
$$

By the continuity of $\mathscr{L}$, and a ULLN applied to $\hat{\mathscr{L}}(\theta)$, we can take $\left\|\hat{\theta} - \theta_\infty\right\|_2$ sufficiently small that that $|\hat{\mathscr{L}}(\hat{\theta}) - \hat{\mathscr{L}}(\theta_\infty)| < \epsilon_L$. It follows that, with probability approaching one,

$$
\frac{1}{N} \sup_{\theta: \left\|\theta - \hat{\theta}\right\|_2 \ge \delta_2} \left(\sum_{n=1}^N \ell(x_n|\hat{\theta}) - \sum_{n=1}^N \ell(x_n|\theta) \right) \le \epsilon_L - 2\epsilon_L = -\epsilon_L.
$$

The result follows by multiplying both sides of the inequality by N. $\square$

Note that Lemma 4 requires δ_2 to be sufficiently small (so that $\delta_{LLN} > \delta_2$), which in turn requires ϵ_U to be small (so that $\delta_{LLN} > 3\epsilon_U$). For any particular $\delta_{LLN} > \delta_2$, we can always apply Lemma 4 with arbitrarily small ϵ_U, at the cost of requiring a large enough N^* so that the probability convergence results apply. Throughout, we will choose ϵ_U *small enough* that we obtain our desired results.

D.4 Region R_3

The integral over R_3 goes to zero by strict optimality of θ_∞ and the proper behavior of the prior.

Lemma 5. *Under Assumption 1, when $\mathfrak{R}$ occurs with $\epsilon_U < \pi(\theta_\infty)/2$,*

$$\|I_3(\phi)\|_2 = O(\exp(-N\epsilon_L)) \underset{\pi(\theta)}{\mathbb{E}} \left[\|\phi(\theta, \mathcal{X}_s)\|_2\right].$$

Proof. We have

$$\begin{aligned}
\|I_3(\phi)\|_2 = &\left\| \int_{R_3} \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\theta \right\|_2 \\
&\leq \sup_{\theta \in R_3} \exp\left(N\left(\frac{1}{N}\sum_{n=1}^{N} \ell(x_n|\theta) - \frac{1}{N}\sum_{n=1}^{N} \ell(x_n|\hat{\theta})\right)\right) \times \int_{R_3} \|\phi(\theta, \mathcal{X}_s)\|_2 \frac{\pi(\theta)}{\pi(\hat{\theta})} d\theta \\
&= \exp(-N\epsilon_L) \frac{1}{\pi(\hat{\theta})} \underset{\pi(\theta)}{\mathbb{E}} \left[\|\phi(\theta, \mathcal{X}_s)\|_2 \mathbb{I}(\theta \in R_3)\right] \quad \text{(Lemma 4, Event 6)} \\
&\leq \exp(-N\epsilon_L) \frac{1}{\pi(\theta_\infty) - \epsilon_U} \underset{\pi(\theta)}{\mathbb{E}} \left[\|\phi(\theta, \mathcal{X}_s)\|_2 \mathbb{I}(\theta \in R_3)\right] \quad \text{(Lemma 4, Event 3)} \\
&\leq \exp(-N\epsilon_L) 2\pi(\theta_\infty)^{-1} \underset{\pi(\theta)}{\mathbb{E}} \left[\|\phi(\theta, \mathcal{X}_s)\|_2\right].
\end{aligned}$$

In the last step, we take $\epsilon_U < \pi(\theta_\infty)/2$ in $\mathfrak{R}$ (specifically, Lemma 4 Event 3). $\square$

We note that the form of the bound in Lemma 5 is designed to take advantage of the prior mean assumption Definition 2 Item 2, or its more general version Definition 2 Item 2.

D.5 Gaussian definitions

For regions R_1 and R_2, and the combination of the regions, we will make use of the following compact notation for key Gaussian densities.

Definition 8. Write the Gaussian density with mean 0 and covariance $\hat{\mathcal{I}}^{-1}$ and its normalizing constant as

$$\hat{\mathcal{G}}(\tau) := \hat{Z}^{-1} \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) = \mathcal{N}(\tau|0, \hat{\mathcal{I}}^{-1}) \quad \text{where} \quad \hat{Z} := \det\left(2\pi\hat{\mathcal{I}}^{-1}\right)^{1/2}.$$

Additionally, let $\mathcal{G}_\lambda^S(\tau)$ denote the $\mathcal{N}\left(\tau|0, \lambda^{-1} I_{D_\theta}\right)$ distribution truncated to the set S, where I_{D_θ} denotes the $D_\theta \times D_\theta$ identity matrix. Specifically,

$$\mathcal{G}_\lambda^S(\tau) := \frac{\exp\left(-\frac{\lambda}{2}\left\|\tau\right\|_2^2\right)}{\int_S \exp\left(-\frac{\lambda}{2}\left\|\tau\right\|_2^2\right) d\tau}.$$

◀

D.6 Region R_2

Region R_2 is dealt with by choosing δ_2 sufficiently small that $I_2(\phi)$ is dominated by a particular Gaussian integral. The we will make that Gaussian integral small by choosing δ_1 sufficiently large.

Lemma 6. *Let Assumption 1 hold. Take $\delta_1 \geq \sqrt{8/\lambda_{\mathcal{I}}}$. There exist δ_2 and ϵ_U, defined as functions only of the population quantities $\lambda_{\mathcal{I}}$ and $\mathscr{L}_{(k)}(\theta)$, such that, for any $\mathcal{X}$ such that $\mathfrak{R}$ occurs,*

$$\left\|I_2(\phi)\right\|_2 \leq O\left(N^{-3}\right) \underset{\mathcal{G}_{\lambda_{\mathcal{I}}/2}^{R_1 \bigcup R_2}(\tau)}{\mathbb{E}} \left[\left\|\phi(\theta, \mathcal{X}_s)\right\|_2\right].$$

Proof. The main part of the proof consists in showing that there exists a δ_2 sufficiently small so that

$$\left\|I_2(\phi)\right\|_2 \leq \int_{R_2} \exp\left(-\frac{\lambda_{\mathcal{I}}}{2}\left\|\tau\right\|_2^2\right) \left\|\phi(\theta, \mathcal{X}_s)\right\|_2 d\tau, \tag{16}$$

The approach will be to bound the difference between $N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})$ and $\frac{1}{2}\mathscr{L}_{(2)}(\theta_\infty)$ so that the entire expression inside the integral's exponent can be written as $A\tau^2$ for a positive definite matrix A.

For the duration of this proof, define the following residual matrix:

$$\begin{aligned}\Lambda(\theta, \mathcal{X}, N) := &\frac{1}{2}\left(\hat{\mathscr{L}}_{(2)}(\hat{\theta}) - \mathscr{L}_{(2)}(\theta_\infty)\right) + \\ &N^{-1/2}\frac{1}{6}\left(\hat{\mathscr{L}}_{(3)}(\hat{\theta}) - \mathscr{L}_{(3)}(\theta_\infty)\right)\tau + \\ &N^{-1/2}\frac{1}{6}\mathscr{L}_{(3)}(\theta_\infty)\tau + N^{-1}\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)\tau^2.\end{aligned}$$

Under this definition, by Eq. 11,

$$N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta}) = \left(\frac{1}{2}\mathscr{L}_{(2)}(\theta_\infty) + \Lambda(\theta, \mathcal{X}, N)\right)\tau^2.$$

This expansion holds for all θ, but we need to control the residual only in R_2. We do so by bounding the entries of the $D_\theta \times D_\theta$ matrix $\Lambda(\theta, \mathcal{X}, N)$. When $\mathfrak{R}$ occurs, we have

$$
\begin{aligned}
\left\|\hat{\mathscr{L}}_{(2)}(\hat{\theta}) - \mathscr{L}_{(2)}(\theta_\infty)\right\|_2 &\leq \left\|\hat{\mathscr{L}}_{(2)}(\hat{\theta}) - \mathscr{L}_{(2)}(\hat{\theta})\right\|_2 + \left\|\mathscr{L}_{(2)}(\hat{\theta}) - \mathscr{L}_{(2)}(\theta_\infty)\right\|_2 \\
&\leq \sup_{\theta \in R_2} \left\|\hat{\mathscr{L}}_{(2)}(\theta) - \mathscr{L}_{(2)}(\theta)\right\|_2 + \sup_{\theta \in R_2} \left\|\mathscr{L}_{(3)}(\theta)\right\|_2 \left\|\hat{\theta} - \theta_\infty\right\|_2 \\
&\leq \left(1 + \sup_{\theta \in R_2} \left\|\mathscr{L}_{(3)}(\theta)\right\|_2\right) \epsilon_U. \quad \text{(Lemma 4, Events 4 and 2)}
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
\left\|\left(\hat{\mathscr{L}}_{(3)}(\hat{\theta}) - \mathscr{L}_{(3)}(\theta_\infty)\right)\tau\right\|_2 &\leq \left\|\hat{\mathscr{L}}_{(3)}(\hat{\theta}) - \mathscr{L}_{(3)}(\theta_\infty)\right\|_2 \|\tau\|_2 \\
&\leq \left(1 + \sup_{\theta \in R_2} \left\|\mathscr{L}_{(4)}(\theta)\right\|_2\right) \|\tau\|_2 \, \epsilon_U.
\end{aligned}
$$

Next, applying Cauchy-Schwarz gives

$$
\left\|\mathscr{L}_{(3)}(\theta_\infty)\tau\right\|_2 \leq \left\|\mathscr{L}_{(3)}(\theta_\infty)\right\|_2 \|\tau\|_2 .
$$

Finally, let us consider the Taylor series residual term. Let $\tilde{\theta}(t) = t(\theta - \hat{\theta}) + \hat{\theta}$. Because we can write

$$
\begin{aligned}
\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right) &= \frac{1}{6}\int_0^1 (t-1)^3 \hat{\mathscr{L}}_{(4)}(\tilde{\theta}(t))dt \\
&= \frac{1}{6}\int_0^1 (t-1)^3 \left(\hat{\mathscr{L}}_{(4)}(\tilde{\theta}(t)) - \mathscr{L}_{(4)}(\tilde{\theta}(t))\right) dt + \frac{1}{6}\int_0^1 (t-1)^3 \mathscr{L}_{(4)}(\tilde{\theta}(t))dt,
\end{aligned}
$$

we have, when $\mathfrak{R}$ occurs,

$$
\begin{aligned}
\sup_{\theta \in R_2} \left\|\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)\tau^2\right\|_2 \leq& \left\|\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}, \hat{\theta}, \theta\right)\right\|_2 \|\tau\|_2^2 \\
\leq& \frac{1}{6}\left(\sup_{\theta \in R_2} \left\|\hat{\mathscr{L}}_{(4)}(\theta) - \mathscr{L}_{(4)}(\theta)\right\|_2 + \sup_{\theta \in R_2} \left\|\mathscr{L}_{(4)}(\theta)\right\|_2\right) \|\tau\|_2^2 \\
\leq& \frac{1}{6}\left(\epsilon_U + \sup_{\theta \in R_2} \left\|\mathscr{L}_{(4)}(\theta)\right\|_2\right) \|\tau\|_2^2 . \quad \text{(Lemma 4, Event 4)}
\end{aligned}
$$

Combining the above results, together with the fact that, in R_2, $\sqrt{N}\left\|\tau\right\|_2 \le \delta_2$, gives

$$
\begin{aligned}
\sup_{\theta\in R_2}\left\|\Lambda(\theta,\mathcal{X},N)\right\|_2 \le & \frac{1}{2}\left\|\hat{\mathscr{L}}_{(2)}(\hat{\theta}) - \mathscr{L}_{(2)}(\theta_\infty)\right\|_2 + \\
& N^{-1/2}\frac{1}{6}\left\|\hat{\mathscr{L}}_{(3)}(\hat{\theta}) - \mathscr{L}_{(3)}(\theta_\infty)\right\| \sup_{\theta\in R_2}\left\|\tau\right\|_2 + \\
& N^{-1/2}\frac{1}{6}\left\|\mathscr{L}_{(3)}(\theta_\infty)\right\|_2 \sup_{\theta\in R_2}\left\|\tau\right\|_2 + \\
& N^{-1}\sup_{\theta\in R_2}\left\|\mathscr{T}_{(4)}\left(\hat{\mathscr{L}},\hat{\theta},\theta\right)\right\|_2 \left\|\tau\right\|_2^2 \\
\le & \frac{1}{2}\left(1 + \sup_{\theta\in R_2}\left\|\mathscr{L}_{(3)}(\theta)\right\|_2\right)\epsilon_U + \frac{1}{6}\left(1 + \sup_{\theta\in R_2}\left\|\mathscr{L}_{(4)}(\theta)\right\|_2\right)\delta_2\epsilon_U + \\
& \frac{1}{6}\left\|\mathscr{L}_{(3)}(\theta_\infty)\right\|_2 \delta_2 + \frac{1}{6}\left(\epsilon_U + \sup_{\theta\in R_2}\left\|\mathscr{L}_{(4)}(\theta)\right\|_2\right)\delta_2^2. \qquad (17)
\end{aligned}
$$

The preceding bound contains only the ϵ_U, δ_2, and the population quantities $\sup_{\theta\in R_2}\left\|\mathscr{L}_{(3)}(\theta)\right\|_2$, $\sup_{\theta\in R_2}\left\|\mathscr{L}_{(4)}(\theta)\right\|_2$, and $\lambda_{\mathcal{I}}$. The quantities $\sup_{\theta\in R_2}\left\|\mathscr{L}_{(3)}(\theta)\right\|_2$ and $\sup_{\theta\in R_2}\left\|\mathscr{L}_{(4)}(\theta)\right\|_2$ depend on δ_2 themselves, but are finite and decreasing in δ_2.

Consequently, when $\mathfrak{R}$ occurs, there exist an ϵ_U and δ_2 sufficiently small so that

$$
\left\|\Lambda(\theta,\mathcal{X},N)\right\|_\infty \le \sqrt{D_\theta^2}\left\|\Lambda(\theta,\mathcal{X},N)\right\|_2 \le \frac{\lambda_{\mathcal{I}}}{2}, \qquad (18)
$$

independently of $\mathcal{X}$ (except that $\mathfrak{R}$ occurs). The bound Eq. 18 cannot necessarily be achieved simply by setting ϵ_U to be small (i.e., by relying on the consistency of $\hat{\theta}$ and the ULLN) because of the terms $\sup_{\theta\in R_2}\left\|\mathscr{L}_{(4)}(\theta)\right\|_2 \delta_2^2$ and $\left\|\mathscr{L}_{(3)}(\theta_\infty)\right\|_2 \delta_2$ in Eq. 17—it is necessary to set δ_2 small as well.

Recall from Definition 1 that $\mathcal{I} = -\frac{1}{2}\mathscr{L}_{(2)}(\theta_\infty)$, and from Definition 4 that $\lambda_{\mathcal{I}} > 0$ is the minimum eigenvalue of $\mathcal{I}$. These facts, combined with Eq. 18 give that, for any τ,

$$
\begin{aligned}
\sup_{\theta\in R_2}\left(\frac{1}{2}\mathscr{L}_{(2)}(\theta_\infty) + \Lambda(\theta,\mathcal{X},N)\right)\tau^2 \le & -\lambda_{\mathcal{I}}\left\|\tau\right\|_2^2 + \frac{\lambda_{\mathcal{I}}}{2}\left\|\tau\right\|_2^2 \\
= & -\frac{\lambda_{\mathcal{I}}}{2}\left\|\tau\right\|_2^2.
\end{aligned}
$$

Plugging into the definition of $I_2(\phi)$ given in Definition 7 gives the desired Eq. 16.

To complete the proof, we use the fact that the normal density is bounded in R_2.

$$
\begin{aligned}
\|I_2(\phi)\|_2 &\leq \int_{R_2} \exp\left(-\frac{\lambda_{\mathcal{I}}}{2}\|\tau\|_2^2\right)\|\phi(\theta,\mathcal{X}_s)\|_2\, d\tau \\
&= \int_{R_2} \exp\left(-\frac{\lambda_{\mathcal{I}}}{4}\|\tau\|_2^2 - \frac{\lambda_{\mathcal{I}}}{4}\|\tau\|_2^2\right)\|\phi(\theta,\mathcal{X}_s)\|_2\, d\tau \\
&\leq \sup_{\tau\in R_2} \exp\left(-\frac{\lambda_{\mathcal{I}}}{4}\|\tau\|_2^2\right)\int_{R_2} \exp\left(-\frac{\lambda_{\mathcal{I}}}{4}\|\tau\|_2^2\right)\|\phi(\theta,\mathcal{X}_s)\|_2\, d\tau \\
&\leq \exp\left(-\frac{\lambda_{\mathcal{I}}}{4}\delta_1^2\log N\right)\int_{R_1\bigcup R_2} \exp\left(-\frac{\lambda_{\mathcal{I}}}{4}\|\tau\|_2^2\right)\|\phi(\theta,\mathcal{X}_s)\|_2\, d\tau.
\end{aligned}
$$

In the final line, we have used the fact that $(\log N)^2 \geq \log N$ for all $N \geq 3$. Taking $\delta_1^2 \geq 12/\lambda_{\mathcal{I}}$ and multiplying by the $O(1)$ normalizer for the $\mathcal{G}_{\lambda_{\mathcal{I}}/2}^{R_1\bigcup R_2}(\tau)$ completes the proof. □

By inspecting the proof of Lemma 6, one can see that that any polynomial rate of convergence can be achieved by taking δ_1 sufficiently large. Indeed, if the lower upper boundary of R_1 were of the form $\tau \leq N^k$ for some $k < 1/2$, then $\|I_2(\phi)\|_2$ would decay exponentially and R_1 would still shrink to a point in the θ domain. However, such a rapidly growing boundary between R_1 and R_2 comes at a cost in the analysis of region R_1. In particular, it is the use of a logarithmically growing boundary is what allows us to give polylog bounds in the subsequent Lemma 10 and in the final result of Theorem 1.

D.7 Region R_1

It is region R_1 that will dominate the value of the integral $I(\phi)$. We will first perform a Taylor series expansion of all the terms around their sample versions, and then take probability limits of the resulting expressions.

First, we will show that integrals over $\hat{\mathcal{G}}(\tau)$ in R_1 are sufficiently close to the corresponding Gaussian integrals over $\mathbb{R}^{D_\theta}$ for sufficiently large δ_1.

Lemma 7. *Let Assumption 1 hold. Let τ^k represent the D_θ-dimensional array of outer products of τ. When $\mathfrak{R}$ occurs and $\delta_1 \geq \sqrt{24/\lambda_{\mathcal{I}}}$ then, for any $k \geq 0$,*

$$
\int_{R_1} \hat{\mathcal{G}}(\tau)\tau^k d\tau = \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)}\left[\tau^k\right] + O(N^{-3}).
$$

Proof. By definition,

$$
\int_{R_1} \hat{\mathcal{G}}(\tau)\tau^k d\tau - \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)}\left[\tau^k\right] = \int_{R_2\bigcup R_3} \hat{\mathcal{G}}(\tau)\tau^k d\tau
$$

We will now use a technique similar to that of the proof of Lemma 6 to control the norm of the final integral. Recall that, on $\mathfrak{R}$, the minimum eigenvalue of $\hat{\mathcal{I}}$, $\hat{\lambda}_{\mathcal{I}}$, is at least $\frac{1}{2}\lambda_{\mathcal{I}}$

(Lemma 4 Event 5), so that $\hat{Z}^{-1} = O(1)$.

$$
\begin{aligned}
&\left\| \int_{R_2 \bigcup R_3} \hat{\mathcal{G}}(\tau)\tau^k d\tau \right\|_2 \\
&\quad \le \int_{R_2 \bigcup R_3} \hat{\mathcal{G}}(\tau) \left\|\tau\right\|_2^k d\tau \\
&\quad = \hat{Z}^{-1} \int_{R_2 \bigcup R_3} \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) \left\|\tau\right\|_2^k d\tau \\
&\quad \le \hat{Z}^{-1} \int_{R_2 \bigcup R_3} \exp\left(-\frac{1}{4}\lambda_{\mathcal{I}} \left\|\tau\right\|_2^2\right) \left\|\tau\right\|_2^k d\tau \\
&\quad \le \sup_{\tau \in R_2 \bigcup R_3} \exp\left(-\frac{1}{8}\lambda_{\mathcal{I}} \left\|\tau\right\|_2^2\right) \hat{Z}^{-1} \int_{R_2 \bigcup R_3} \exp\left(-\frac{1}{8}\lambda_{\mathcal{I}} \left\|\tau\right\|_2^2\right) \left\|\tau\right\|_2^k d\tau \\
&\quad = \exp\left(-\frac{1}{8}\lambda_{\mathcal{I}}\delta_1^2 (\log N)^2\right) \hat{Z}^{-1} \int_{R_2 \bigcup R_3} \exp\left(-\frac{1}{8}\lambda_{\mathcal{I}} \left\|\tau\right\|_2^2\right) \left\|\tau\right\|_2^k d\tau \\
&\quad = O(N^{-3}),
\end{aligned}
$$

where the final line takes $\delta_1^2 \ge \frac{24}{\lambda_{\mathcal{I}}}$, that $\log N < (\log N)^2$ for $N \ge 3$, that $\hat{Z}^{-1} = O(1)$, and that normal moments are finite. □

Next, we series expand the likelihood term.

Lemma 8. *Let Assumption 1 hold. For any $\mathcal{X}$ such that $\mathfrak{R}$ occurs, and for $\theta \in R_1$,*

$$
\begin{aligned}
&\exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) = \\
&\qquad \exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) \times \left(1 + N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\theta)\tau^3 + \tilde{O}\left(N^{-1}\right)\right).
\end{aligned}
$$

Proof. Using Eq. 11, write the log likelihood term as

$$
\begin{aligned}
&\exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) = \\
&\qquad \exp\left(\frac{1}{2}\hat{\mathscr{L}}_{(2)}(\hat{\theta})\tau^2\right) \times \\
&\qquad \exp\left(N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^3 + N^{-1}\frac{1}{6}\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}(\theta), \hat{\theta}, \theta\right)\tau^4\right).
\end{aligned}
\tag{19}
$$

We will expand the second term of Eq. 19 using the expansion of the exponential function around 0, which holds for all z:

$$
\begin{aligned}
\exp(z) &= 1 + z + z^2 + \frac{1}{2}\int_0^1 (1-t)^2 \exp(tz) dt z^3 \\
&= 1 + z + O(|z|^2).
\end{aligned}
$$

To match Eq. 19, we need to evaluate at

$$z = N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^3 + N^{-1}\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}(\theta), \hat{\theta}, \theta\right)\tau^4.$$

When $\mathfrak{R}$ occurs (Lemma 4 Event 4), we have

$$\begin{aligned}
\left\|N^{-1}\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}(\theta), \hat{\theta}, \theta\right)\tau^4\right\|_2 \leq & N^{-1}\sup_{\theta\in R_1}\left\|\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}(\theta), \hat{\theta}, \theta\right)\right\|_2 \sup_{\tau\in R_1}\left\|\tau^3\right\|_2 \\
= & \tilde{O}\left(N^{-1}\right) \\
\left\|N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^3\right\|_2 \leq & N^{-1/2}\frac{1}{6}\left\|\hat{\mathscr{L}}_{(3)}(\hat{\theta})\right\|_2 \sup_{\tau\in R_1}\left\|\tau^3\right\|_2 \\
= & \tilde{O}\left(N^{-1/2}\right) \Rightarrow \\
|z| = & \tilde{O}\left(N^{-1/2}\right),
\end{aligned}$$

so that

$$\begin{aligned}
\exp\left(N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^3 + N^{-1}\frac{1}{6}\mathscr{T}_{(4)}\left(\hat{\mathscr{L}}(\theta), \hat{\theta}, \theta\right)\tau^4\right) = & \\
1 + N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\theta)\tau^3 + \tilde{O}\left(N^{-1}\right). & \qquad (20)
\end{aligned}$$

Combining Eqs. 19 and 20 and recognizing the definition of $\hat{\mathcal{I}}$ gives the desired result. □

Lemma 9. *Let Assumption 1 hold. When $\mathfrak{R}$ occurs and δ_1 is as given in Lemma 7,*

$$\int_{R_1}\exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right)d\tau = \hat{Z} + \tilde{O}\left(N^{-1}\right).$$

Proof. Recall that, by definition,

$$\hat{Z} = \int\exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)d\tau.$$

Expanding using Lemma 8 gives

$$\begin{aligned}
&\left|\int_{R_1} \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau - \hat{Z}\right| \\
&= \left|\int_{R_1} \exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)\left(1 + N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\theta)\tau^3 + \tilde{O}\left(N^{-1}\right)\right) d\tau - \int \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) d\tau\right| \\
&= \left|\int_{R_1} \exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)\left(N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\theta)\tau^3 + \tilde{O}\left(N^{-1}\right)\right) d\tau - \int_{R_2 \bigcup R_3} \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) d\tau\right| \\
&\le \frac{1}{6}N^{-1/2} \sup_{\theta \in R_1} \left\|\hat{\mathscr{L}}_{(3)}(\theta)\right\|_2 \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)}\left[\mathbb{I}\left(\tau \in R_1\right)\tau^3\right] + \sup_{\tau \in R_2 \bigcup R_3} \exp\left(-\frac{1}{4}\lambda_{\mathcal{I}}\left\|\tau\right\|_2^2\right) + \tilde{O}\left(N^{-1}\right) \\
&= O(N^{-2}) + \exp\left(-\frac{1}{4}\lambda_{\mathcal{I}}\delta_1^2(\log N)^2\right) + \tilde{O}\left(N^{-1}\right) \\
&= \tilde{O}\left(N^{-1}\right),
\end{aligned}$$

where, in the penultimate line, we use Lemma 7, including the given lower bound on δ_1. □

In Lemma 10 to follow, we integrate the result of Lemma 8 using Lemma 7 to produce our final results for region R_1. Recall from Definition 4 that $\hat{\mathcal{G}}(\tau) = \hat{Z}^{-1}\exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)$ is the density of the Gaussian distribution on τ with mean zero and covariance $\hat{\mathcal{I}}^{-1} = (-\hat{\mathscr{L}}_{(2)}(\hat{\theta}))^{-1}$.

Note that the statement of Lemma 10 combines vector-valued quantities like $\phi_{(2)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)}\left[\tau^2\right]$ with scalar–valued quantities like $\left\|\phi_{(2)}(\hat{\theta}, \mathcal{X}_s)\right\|_2$. This is for notational convenience since the scalar-valued terms will simply become part of the residual, and their component values do not matter. Formally, the equation can be taken to hold componentwise.

Lemma 10. *Let Assumption 1 hold. When δ_1 is as given in Lemma 6, then, for any $\mathcal{X}$ such that $\mathfrak{R}$ occurs,*

$$\begin{aligned}
I_1(\phi) = \phi(\hat{\theta}, \mathcal{X}_s) &\int_{R_1} \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau + \\
&N^{-1}\hat{Z}\left(\frac{1}{2}\phi_{(2)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)}\left[\tau^2\right] + \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\phi_{(1)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)}\left[\tau^4\right]\right) + \\
&\tilde{O}\left(N^{-2}\right)\left(1 + \left\|\phi_{(1)}(\hat{\theta}, \mathcal{X}_s)\right\|_2 + \left\|\phi_{(2)}(\hat{\theta}, \mathcal{X}_s)\right\|_2 + \mathop{\mathbb{E}}_{\mathcal{G}^{R_1}_{\lambda_{\mathcal{I}}/2}(\tau)}\left[\left\|\mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)\right\|_2\right]\right).
\end{aligned}$$

Proof. The result will follow by Taylor expanding ϕ, combining with Lemma 8, and then integrating using the fact that Gaussian integrals in region R_1 are approximately equal to Gaussian integrals over the whole domain, as proved in Lemma 7.

We first Taylor expand $\phi(\theta, \mathcal{X}_s)$ around $\hat{\theta}$ and substitute $\tau = \sqrt{N}(\theta - \hat{\theta})$.

$$\begin{aligned}
\phi(\theta, \mathcal{X}_s) = \phi(\hat{\theta}, \mathcal{X}_s) + N^{-1/2}\phi_{(1)}(\hat{\theta}, \mathcal{X}_s)\tau + \\
N^{-1}\frac{1}{2}\phi_{(2)}(\hat{\theta}, \mathcal{X}_s)\tau^2 + N^{-3/2}\mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)\tau^3.
\end{aligned} \tag{21}$$

We will now combine Eq. 21 with Lemma 8.

$$
\begin{aligned}
&\exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right)\left(\phi(\theta,\mathcal{X}_s) - \phi(\hat{\theta},\mathcal{X}_s)\right) = \\
&\exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)\times \\
&\quad\left(N^{-1/2}\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau + N^{-1}\frac{1}{2}\phi_{(2)}(\hat{\theta},\mathcal{X}_s)\tau^2 + N^{-3/2}\mathscr{T}_{(3)}\left(\phi(\cdot,\mathcal{X}_s),\theta,\hat{\theta}\right)\tau^3\right)\times \\
&\quad\left(1 + N^{-1/2}\frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\tau^3 + \tilde{O}\left(N^{-1}\right)\right) = \\
&\exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)\times \\
&\quad\Bigg(N^{-1/2}\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau + \\
&\qquad N^{-1}\left(\frac{1}{2}\phi_{(2)}(\hat{\theta},\mathcal{X}_s)\tau^2 + \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau^4\right) + \\
&\qquad N^{-3/2}\left(\mathscr{T}_{(3)}\left(\phi(\cdot,\mathcal{X}_s),\theta,\hat{\theta}\right)\tau^3 + \frac{1}{12}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\phi_{(2)}(\hat{\theta},\mathcal{X}_s)\tau^5\right) + \\
&\qquad \tilde{O}\left(N^{-3/2}\right)\left(\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau + N^{-1/2}\frac{1}{2}\phi_{(2)}(\hat{\theta},\mathcal{X}_s)\tau^2 + N^{-1}\mathscr{T}_{(3)}\left(\phi(\cdot,\mathcal{X}_s),\theta,\hat{\theta}\right)\tau^3\right)\Bigg) = \\
&\exp\left(\frac{1}{2}\hat{\mathcal{I}}\tau^2\right)\times \\
&\quad\Bigg(N^{-1/2}\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau + N^{-1}\left(\frac{1}{2}\phi_{(2)}(\hat{\theta},\mathcal{X}_s)\tau^2 + \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau^4\right) + \\
&\qquad \tilde{O}\left(N^{-3/2}\right)\Bigg(\phi_{(1)}(\hat{\theta},\mathcal{X}_s)\tau + \left(O\left(1\right)\tau^5 + O\left(N^{-1/2}\right)\tau^2\right)\phi_{(2)}(\hat{\theta},\mathcal{X}_s) + \\
&\qquad\qquad O\left(1\right)\mathscr{T}_{(3)}\left(\phi(\cdot,\mathcal{X}_s),\theta,\hat{\theta}\right)\tau^3\Bigg)\Bigg). \qquad (22)
\end{aligned}
$$

In the final line of Eq. 22 we absorbed some higher-order quantities into the lower-order $\tilde{O}\left(N^{-3/2}\right)$ term for convenience of expression, and used the fact that, when $\mathfrak{R}$ occurs, for $k = 1, 2, 3$, $\left\|\hat{\mathscr{L}}_{(k)}(\hat{\theta})\right\|_2 = O\left(1\right)$ when $\mathfrak{R}$ occurs, since

$$
\left\|\hat{\mathscr{L}}_{(k)}(\hat{\theta}) - \mathscr{L}_{(k)}(\hat{\theta})\right\|_2 \le \sup_{\theta\in R_1}\left\|\hat{\mathscr{L}}_{(k)}(\theta) - \mathscr{L}_{(k)}(\theta)\right\|_2 \le \epsilon_U,
$$

so

$$
\begin{aligned}
\left\|\hat{\mathscr{L}}_{(k)}(\hat{\theta})\right\|_2 \le& \left\|\hat{\mathscr{L}}_{(k)}(\hat{\theta}) - \mathscr{L}_{(k)}(\hat{\theta})\right\|_2 + \left\|\mathscr{L}_{(k)}(\hat{\theta}) - \mathscr{L}_{(k)}(\theta_\infty)\right\|_2 + \left\|\mathscr{L}_{(k)}(\theta_\infty)\right\|_2 \\
\le&\epsilon_U + \sup_{\theta\in R_1}\left\|\mathscr{L}_{(k)}(\theta) - \mathscr{L}_{(k)}(\theta_\infty)\right\|_2 + \left\|\mathscr{L}_{(k)}(\theta_\infty)\right\|_2 = O\left(1\right),
\end{aligned}
$$

by continuity of $\mathscr{L}_{(k)}(\theta)$.

Next, we integrate both sides of Eq. 22 using Lemma 7 to replace Gaussian integrals over R_1 with Gaussian moments over the whole domain, up to an order N^{-3} which can be absorbed in the $\tilde{O}\left(N^{-2}\right)$ error term.

$$\begin{aligned}
I_1(\phi) = & \int_{R_1} \phi(\theta, \mathcal{X}_s) \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau \\
= & \phi(\hat{\theta}, \mathcal{X}_s) \int_{R_1} \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat{\theta})\right) d\tau + \\
& N^{-1}\left(\frac{1}{2}\phi_{(2)}(\hat{\theta}, \mathcal{X}_s) \underset{\hat{\mathcal{G}}(\tau)}{\mathbb{E}}\left[\tau^2\right] + \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat{\theta})\phi_{(1)}(\hat{\theta}, \mathcal{X}_s) \underset{\hat{\mathcal{G}}(\tau)}{\mathbb{E}}\left[\tau^4\right]\right) + \\
& \tilde{O}\left(N^{-2}\right)\left(1 + \left\|\phi_{(1)}(\hat{\theta}, \mathcal{X}_s)\right\|_2 + \left\|\phi_{(2)}(\hat{\theta}, \mathcal{X}_s)\right\|_2 + \right. \\
& \qquad \left. \int_{R_1} \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) \mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right) d\tau\right).
\end{aligned}$$

Finally, since $\mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)$ depends explicitly on τ, we must control this term in a more convenient form. We will argue as in Lemma 7. Recall that, when $\mathfrak{R}$ occurs, $\hat{\lambda}_{\mathcal{I}} \geq \frac{\lambda_{\mathcal{I}}}{2}$, so that $\hat{Z} = O(1)$ and

$$\begin{aligned}
& \left\|\hat{Z}\int_{R_1} \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) \mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right) d\tau\right\|_2 \\
\leq & \hat{Z}\int_{R_1} \exp\left(-\frac{1}{2}\hat{\mathcal{I}}\tau^2\right) \left\|\mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)\right\|_2 d\tau \\
\leq & \hat{Z}\int_{R_1} \exp\left(-\frac{\lambda_{\mathcal{I}}}{4}\left\|\tau\right\|_2^2\right) \left\|\mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)\right\|_2 d\tau.
\end{aligned}$$

The result follows by multiplying by the $O(1)$ normalizer of the $\mathcal{G}^{R_1}_{\lambda_{\mathcal{I}}/2}$ distribution. □

D.8 Combining the regions

We now combine the results of the previous three sections to evaluate $I(\phi)$. The accuracy of our BCLT expansion of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\phi(\theta, \mathcal{X}_s)\right]$ will depend on the following residual function.

Definition 9. Let $\lambda_{\mathcal{I}} > 0$ denote the minimum eigenvalue of $\mathcal{I}$, and let $\delta > 0$ be an arbitrarily small number. For a function $\phi(\theta, \mathcal{X}_s)$ that is third-order BCLT-okay, define the residual function

$$
\begin{aligned}
\mathscr{R}^0(\mathcal{X}_s) &:= 1 + \left\|\phi(\hat\theta, \mathcal{X}_s)\right\|_2 + \left\|\phi_{(1)}(\hat\theta, \mathcal{X}_s)\right\|_2 + \left\|\phi_{(2)}(\hat\theta, \mathcal{X}_s)\right\|_2 \\
\mathscr{R}^1(\mathcal{X}_s) &:= \underset{\mathcal{G}^{R_1}_{\lambda_{\mathcal{I}}/2}(\tau)}{\mathbb{E}} \left[\left\|\mathscr{T}_{(3)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat\theta\right)\right\|_2\right] \\
\mathscr{R}^2(\mathcal{X}_s) &:= \underset{\mathcal{G}^{R_1 \bigcup R_2}_{\lambda_{\mathcal{I}}/2}(\tau)}{\mathbb{E}} \left[\|\phi(\theta, \mathcal{X}_s)\|_2\right] \\
\mathscr{R}^3(\mathcal{X}_s) &:= \underset{\pi(\theta)}{\mathbb{E}} \left[\|\phi(\theta, \mathcal{X}_s)\|_2\right] \\
\mathscr{R}^\phi(\mathcal{X}_s) &:= \mathscr{R}^0(\mathcal{X}_s) + \mathscr{R}^1(\mathcal{X}_s) + \mathscr{R}^2(\mathcal{X}_s) + \mathscr{R}^3(\mathcal{X}_s).
\end{aligned}
$$

◀

Lemma 11. *Let Assumption 1 hold, and assume that ϕ is third-order BCLT-okay. With $\mathbb{F}$-probability approaching one,*

$$
\begin{aligned}
&\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}} \left[\phi(\theta, \mathcal{X}_s)\right] - \phi(\hat\theta, \mathcal{X}_s) = \\
&\qquad N^{-1}\left(\frac{1}{2}\phi_{(2)}(\hat\theta, \mathcal{X}_s) \underset{\hat{\mathcal{G}}(\tau)}{\mathbb{E}}\left[\tau^2\right] + \frac{1}{6}\mathscr{L}_{(3)}(\hat\theta)\phi_{(1)}(\hat\theta, \mathcal{X}_s) \underset{\hat{\mathcal{G}}(\tau)}{\mathbb{E}}\left[\tau^4\right]\right) + \\
&\qquad \tilde{O}\left(N^{-2}\right)\mathscr{R}^\phi(\mathcal{X}_s),
\end{aligned}
$$

where $\mathscr{R}^\phi(\mathcal{X}_s)$ is given in Definition 9.

Proof. First, combine the results of Lemmas 5, 6 and 10, consolidating higher orders into lower orders and expanding the domain of integration in the residual $\mathscr{R}^\phi(\mathcal{X}_s)$.

$$
\begin{aligned}
I(\phi) = \phi(\hat\theta, \mathcal{X}_s) &\int_{R_1} \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat\theta)\right) d\tau + \\
&N^{-1}\hat{Z}\left(\frac{1}{2}\phi_{(2)}(\hat\theta, \mathcal{X}_s) \underset{\hat{\mathcal{G}}(\tau)}{\mathbb{E}}\left[\tau^2\right] + \frac{1}{6}\hat{\mathscr{L}}_{(3)}(\hat\theta)\phi_{(1)}(\hat\theta, \mathcal{X}_s) \underset{\hat{\mathcal{G}}(\tau)}{\mathbb{E}}\left[\tau^4\right]\right) + \\
&\tilde{O}\left(N^{-2}\right)\mathscr{R}^\phi(\mathcal{X}_s). \qquad (23)
\end{aligned}
$$

For convenience in the final bound of Theorem 1 we have included the term $\left\|\phi(\hat\theta, \mathcal{X}_s)\right\|_2$ in the definition of the residual even though it does not occur in Lemma 10.

The final expectation is given by the ratio

$$
\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}} \left[\phi(\theta, \mathcal{X}_s)\right] = \frac{I(\phi)}{I(1)}.
$$

For the remainder of this proof, for compactness of notation, we define

$$
\begin{aligned}
\zeta &:= \int_{R_1} \exp\left(N\hat{\mathscr{L}}(\theta) - N\hat{\mathscr{L}}(\hat\theta)\right) d\tau \\
&= \hat{Z} + \tilde{O}\left(N^{-1}\right) \\
&= \hat{Z}(1 + \tilde{O}\left(N^{-1}\right)),
\end{aligned}
$$

which follows from Lemma 9 and the fact that $\hat{Z}$ is bounded away from zero when $\mathfrak{R}$ occurs. When $\phi(\theta, \mathcal{X}_s) \equiv 1$, then $\mathscr{R}^0(\mathcal{X}_s) = \mathscr{R}^1(\mathcal{X}_s) = 0$, so by Lemmas 5 and 6,

$$I(1) = \zeta \left(1 + O\left(N^{-3}\right)\right).$$

We now use the expansion, valid for $|z| < 1$ as $z \to 0$,

$$\frac{1}{1+z} = 1 - z + O(z^2). \tag{24}$$

Using Eq. 23 for $I(\phi)$ and applying Eq. 24 to $1/I(1)$ and to $1/\zeta$ gives

$$\begin{aligned}
\frac{I(\phi)}{I(1)} &= \frac{I(\phi)}{\zeta(1 + O\left(N^{-3}\right))} \\
&= \frac{I(\phi)}{\zeta} + O\left(N^{-3}\right) I(\phi) \\
&= \phi(\hat{\theta}, \mathcal{X}_s) + N^{-1} \frac{\hat{Z}}{\zeta} \left(\frac{1}{2} \phi_{(2)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)} \left[\tau^2\right] + \frac{1}{6} \hat{\mathscr{L}}_{(3)}(\hat{\theta}) \phi_{(1)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)} \left[\tau^4\right] \right) + \\
&\quad \tilde{O}\left(N^{-2}\right) \mathscr{R}^{\phi}(\mathcal{X}_s) \\
&= \phi(\hat{\theta}, \mathcal{X}_s) + N^{-1} \left(\frac{1}{2} \phi_{(2)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)} \left[\tau^2\right] + \frac{1}{6} \hat{\mathscr{L}}_{(3)}(\hat{\theta}) \phi_{(1)}(\hat{\theta}, \mathcal{X}_s) \mathop{\mathbb{E}}_{\hat{\mathcal{G}}(\tau)} \left[\tau^4\right] \right) + \\
&\quad \tilde{O}\left(N^{-2}\right) \mathscr{R}^{\phi}(\mathcal{X}_s).
\end{aligned}$$

Note that we have absorbed terms from the expansion of $I(\phi)$ that are linear in $|\phi(\hat{\theta}, \mathcal{X}_s)|$, $\left\|\phi_{(2)}(\hat{\theta}, \mathcal{X}_s)\right\|$, and $\left\|\phi_{(3)}(\hat{\theta}, \mathcal{X}_s)\right\|$ into the residual $\mathscr{R}^{\phi}(\mathcal{X}_s)$. To do so we have used the fact that, when $\mathfrak{R}$ occurs, $\hat{\mathscr{L}}_{(3)}(\hat{\theta}) = \mathscr{L}_{(3)}(\theta_\infty) + O(1)$.

Finally, the expansion holds with $\mathbb{F}$-probability approaching one by Lemma 4. □

D.9 Proof that the BCLT residual is square-summable

All that remains to prove Theorem 1 is to show that the residual is $O_p(1)$ when summed. The form of $\mathscr{R}^{\phi}(\mathcal{X}_s)$ is awkward, but it is designed specifically to commute with suprema over sums, allowing us to apply ULLNs to the remainder of our BCLT, as stated formally in Lemmas 12 and 13 below.

Observe that residual expressions of the form $\sup_{\theta \in R_1 \bigcup R_2} \|\phi(\theta, x)\|_2$ would not be as useful, since

$$\frac{1}{N} \sum_{n=1}^{N} \sup_{\theta \in R_1 \bigcup R_2} \|\phi(\theta, x_n)\|_2 \geq \sup_{\theta \in R_1 \bigcup R_2} \frac{1}{N} \sum_{n=1}^{N} \|\phi(\theta, x_n)\|_2,$$

and ULLN bound on the latter does not imply a bound on the former, since the supremum may depend on x_n in a way that causes the sample mean to diverge.

Lemma 12. *The Gaussian integrals of Definition 9 commutes with sums. For a scalar–valued* *function $\phi(\theta, \mathcal{X}_s)$ and an index set $\mathcal{S}$,*

$$\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \underset{\mathcal{G}_{\lambda_{\mathcal{I}}/2}^{R_1\bigcup R_2}(\tau)}{\mathbb{E}} \left[\phi(\theta, \mathcal{X}_s)\right]^2 \leq \sup_{\theta\in R_1\bigcup R_2} \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \phi(\theta, \mathcal{X}_s)^2.$$

Proof.

$$\begin{aligned}
\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \underset{\mathcal{G}_{\lambda_{\mathcal{I}}/2}^{R_1\bigcup R_2}(\tau)}{\mathbb{E}} \left[\phi(\theta, \mathcal{X}_s)\right]^2 &\leq \underset{\mathcal{G}_{\lambda_{\mathcal{I}}/2}^{R_1\bigcup R_2}(\tau)}{\mathbb{E}} \left[\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \phi(\theta, \mathcal{X}_s)^2\right] \\
&\leq \sup_{\tau\in R_1\bigcup R_2} \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \phi(\hat{\theta}+\tau/\sqrt{N}, \mathcal{X}_s)^2 \\
&= \sup_{\theta\in R_1\bigcup R_2} \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \phi(\theta, \mathcal{X}_s)^2.
\end{aligned}$$

□

Lemma 13. *The Taylor series residual of Definition 9 commutes with sums. When $\theta \in R_1 \bigcup R_2$, for a k-th order continuously differentiable function $\phi(\theta, \mathcal{X}_s)$ and an index set $\mathcal{S}$,*

$$\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \left\|\mathscr{T}_{(k)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)\right\|_2^2 \leq \sup_{\theta\in R_1\bigcup R_2} \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \left\|\phi_{(k)}(\theta, \mathcal{X}_s)\right\|_2^2.$$

Proof. Recall from Definition 3 that

$$\begin{aligned}
&\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \left\|\mathscr{T}_{(k)}\left(\phi(\cdot, \mathcal{X}_s), \theta, \hat{\theta}\right)\right\|_2^2 = \\
&\quad \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \left\|\frac{1}{(k-1)!}\int_0^1 (1-t)^{k-1}\phi_{(k)}(\hat{\theta}+t(\theta-\hat{\theta}), \mathcal{X}_s)dt\right\|_2^2 \leq \\
&\quad \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \frac{1}{(k-1)!}\int_0^1 \left\|\phi_{(k)}(\hat{\theta}+t(\theta-\hat{\theta}), \mathcal{X}_s)\right\|_2^2 dt = \\
&\quad \frac{1}{(k-1)!}\int_0^1 \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \left\|\phi_{(k)}(\hat{\theta}+t(\theta-\hat{\theta}), \mathcal{X}_s)\right\|_2^2 dt \leq \\
&\quad \frac{1}{(k-1)!}\sup_{\theta\in R_1\bigcup R_2} \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \left\|\phi_{(k)}(\theta, \mathcal{X}_s)\right\|_2^2.
\end{aligned}$$

The result follows because $k \geq 1$. □

Lemma 14. *Under Assumption 1, $\frac{1}{N}\sum_{n=1}^{N} \mathscr{R}^{\phi}(x_n)^2 = O_p(1)$.*

Proof. We may take δ_2 to be the smaller of the values required for Definition 2 Item 3 and the region R_2 Lemma 6 to hold. Then, by Definition 2 Item 3, for $k = 0, 1, 2, 3$,

$$\sup_{\theta \in R_1 \bigcup R_2} \frac{1}{N} \sum_{n=1}^{N} \left\| \phi_{(k)}(\theta, x_n) \right\|_2^2 = O_p(1),$$

It follows immediately that $\frac{1}{N} \sum_{n=1}^{N} \left\| \phi_{(k)}(\hat{\theta}, x_n) \right\|_2 = O_p(1)$ and $\frac{1}{N} \sum_{n=1}^{N} \left\| \phi_{(k)}(\hat{\theta}, x_n) \right\|_2^2 = O_p(1)$ for $k = 0, 1, 2, 3$. From this and Cauchy-Schwarz it follows that $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^0(x_n)^2 = O_p(1)$.

Applying Lemma 12 with $\mathcal{S} = \{1, \ldots, N\}$ and $\phi(\theta, \mathcal{X}_s) = \|\phi(\theta, x_n)\|_2$ gives $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^2(x_n) = O_p(1)$ and $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^2(x_n)^2 = O_p(1)$.

Similarly, $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^1(x_n)^2 = O_p(1)$ follows from Lemma 12 and then Lemma 13.

The terms $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^3(x_n)$ and $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^3(x_n)^2$ are controlled by assumption in Definition 2 Item 2.

Finally, the fact that $\frac{1}{N} \sum_{n=1}^{N} \mathscr{R}^\phi(x_n)^2 = O_p(1)$ follows by Cauchy-Schwarz, since $\mathscr{R}^\phi(x_n)$ is the sum of products of terms, the squares of each of which have $O_p(1)$ sum. □

E Proof of Theorem 2

Proof. We can re-arrange the covariance in the definition of ψ_n as the product of terms centered at $\hat{\theta}$:

$$\begin{aligned}\psi_n = N \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})} \left[(g(\theta) - g(\hat{\theta}))(\ell(x_n|\theta) - \ell(x_n|\hat{\theta})) \right] - \\ 2N \left(g(\hat{\theta}) - \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})} [g(\theta)] \right) \left(\ell(x_n|\hat{\theta}) - \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})} [\ell(x_n|\theta)] \right).\end{aligned}$$

Apply Theorem 1 three times, with $\phi_g(\theta) := g(\theta)$, $\phi_\ell(\theta, x_n) := \ell(x_n|\theta)$, and $\phi_{\ell g}(\theta, x_n) := (g(\theta) - g(\hat{\theta}))(\ell(x_n|\theta) - \ell(x_n|\hat{\theta}))$, all of which are third-order BCLT-okay by assumption. Let the residuals be denoted as $\mathscr{R}^g$ (with no x_n dependence), $\mathscr{R}^\ell(x_n)$, and $\mathscr{R}^{\ell g}(x_n)$, respectively. Note that the first derivative of $\theta \mapsto \phi_{\ell g}(\theta, x_n)$ is zero at $\hat{\theta}$.

Write the leading term in the expansion of $\phi_\ell(\theta, x_n)$ as

$$L(x_n) := \frac{1}{2} \ell_{(2)}(\hat{\theta}, x_n) \hat{\mathcal{I}}^{-1} + \frac{1}{6} \ell_{(1)}(\hat{\theta}, x_n) \hat{\mathscr{L}}_{(3)}(\hat{\theta}) \hat{\mathcal{M}}.$$

Under Assumption 1, $\hat{\theta} \xrightarrow[N \to \infty]{prob} \theta_\infty$ and $\hat{\mathcal{I}}^{-1}$ has bounded operator norm with probability approaching one (Lemma 3). These facts imply that $\frac{1}{N} \sum_{n=1}^{N} \|L(x_n)\|_2^2 = O_p(1)$ by Cauchy-Schwarz and a ULLN (Assumption 1 Item 5 and Lemma 2).

Then Theorem 1, together with the fact that $\hat{\theta} = O_p(1)$ and $g(\theta)$ is continuous, gives

$$
\begin{aligned}
\psi_n = & g_{(1)}(\hat{\theta})\hat{\mathcal{I}}\ell_{(1)}(x_n|\hat{\theta}) + \tilde{O}\left(N^{-1}\right)\mathscr{R}^{\ell g}(x_n) + \\
& N\left(O_p\left(N^{-1}\right)L(x_n) + \tilde{O}_p\left(N^{-2}\right)\mathscr{R}^{\ell}(x_n)\right)\left(O_p\left(N^{-1}\right) + \tilde{O}_p\left(N^{-2}\right)\mathscr{R}^{g}\right) \\
= & g_{(1)}(\hat{\theta})\hat{\mathcal{I}}\ell_{(1)}(x_n|\hat{\theta}) + \\
& \tilde{O}\left(N^{-1}\right)\left(\mathscr{R}^{\ell g}(x_n) + L(x_n) + \mathscr{R}^{g}L(x_n) + \mathscr{R}^{\ell}(x_n) + \mathscr{R}^{\ell}(x_n)\mathscr{R}^{g}\right).
\end{aligned}
$$

In the final line, we have combined lower orders into the $O_p(N^{-1})$ term for conciseness of expression.

We now show that the sums of the squares of the residuals in the preceding display go to zero. By Theorem 1, each of $\frac{1}{N}\sum_{n=1}^{N}\mathscr{R}^{\ell}(x_n)^2$, and $\frac{1}{N}\sum_{n=1}^{N}\mathscr{R}^{\ell g}(x_n)^2$ are $O_p(1)$. Similarly, $\frac{1}{N}\sum_{n=1}^{N}\left\|\hat{\mathcal{I}}^{-1}\ell_{(1)}(x_n|\hat{\theta})\right\|_2^2 = O_p(1)$ by a ULLN and the boundedness of the operator norm of $\hat{\mathcal{I}}^{-1}$. By Cauchy-Schwarz, we then have $\frac{1}{N}\sum_{n=1}^{N} g_{(1)}(\hat{\theta})^\intercal\hat{\mathcal{I}}^{-1}\ell_{(1)}(x_n|\hat{\theta})\mathscr{R}^{\psi}(x_n) = O_p(1)$.

By repeated application of Cauchy-Schwarz, we can thus sum ψ_n and $\psi_n\psi_n^\intercal$ to get

$$
\begin{aligned}
\sum_{n=1}^{N}\psi_n &= g_{(1)}(\hat{\theta})\hat{\mathcal{I}}\sum_{n=1}^{N}\ell_{(1)}(x_n|\hat{\theta}) + \tilde{O}_p\left(N^{-1}\right) = \tilde{O}_p\left(N^{-1}\right) \\
\frac{1}{N}\sum_{n=1}^{N}\psi_n\psi_n^\intercal &= g_{(1)}(\hat{\theta})\hat{\mathcal{I}}^{-1}\hat{\Sigma}\hat{\mathcal{I}}^{-1}g_{(1)}(\hat{\theta})^\intercal + \tilde{O}_p\left(N^{-1}\right) = \hat{V}^{\mathrm{Sand}} + \tilde{O}_p\left(N^{-1}\right).
\end{aligned}
$$

Consistency of V^{IJ} then follows from Corollary 1. □

F Weighting representation

To derive Eq. 10, we first derive a general formula for the second derivative with respect to the data weights $\underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[g(\theta)]$, and then use the Tower property on the resulting expression. For compactness, let $\ell_n(\theta) = \ell(x_n|\theta)$. Recall that Eq. 9 is a special case of the general formula

$$
\begin{aligned}
\left.\frac{\partial \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)]}{\partial w_n}\right|_{\mathcal{W}} &= \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathrm{Cov}}\left(a(\theta), \ell_n(\theta)\right) \\
&= \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_n(\theta)] - \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)]\underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)],
\end{aligned}
$$

which we can apply recursively to itself using the chain rule as follows.

$$
\begin{aligned}
&\left.\frac{\partial^2 \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)]}{\partial w_n \partial w_m}\right|_{\mathcal{W}} \\
&= \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathrm{Cov}}(a(\theta)\ell_n(\theta), \ell_m(\theta)) - \\
&\quad \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathrm{Cov}}(a(\theta), \ell_m(\theta)) \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)] - \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathrm{Cov}}(\ell_n(\theta), \ell_m(\theta)) \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)] \\
&= \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_n(\theta)\ell_m(\theta)] - \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_n(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_m(\theta)] - \\
&\quad \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_m(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)] + \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_m(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)] - \\
&\quad \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)\ell_m(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)] + \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_m(\theta)]. \\
&= \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_n(\theta)\ell_m(\theta)] + 2 \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_m(\theta)] - \\
&\quad \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_n(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_m(\theta)] - \\
&\quad \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)\ell_m(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)] - \\
&\quad \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)\ell_m(\theta)] \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)] \\
&= \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}\left[(a(\theta) - \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[a(\theta)])(\ell_n(\theta) - \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_n(\theta)])(\ell_m(\theta) - \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\ell_m(\theta)])\right].
\end{aligned}
$$

Now suppose that $\theta = (\gamma, \lambda)$, and that $a(\theta) = \gamma$. By applying the Tower property to the second derivative,

$$
\left.\frac{\partial^2 \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\gamma]}{\partial^2 w_n}\right|_{1_N} = \underset{\mathbb{P}(\gamma|\mathcal{X})}{\mathbb{E}}\left[(\gamma - \underset{\mathbb{P}(\gamma|\mathcal{X})}{\mathbb{E}}[\gamma]) \underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}\left[(\ell_n(\gamma,\lambda) - \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\ell_n(\gamma,\lambda)])^2\right]\right].
$$

Note that

$$
\begin{aligned}
&\underset{\mathbb{P}(\lambda|\gamma,\mathcal{X})}{\mathbb{E}}\left[(\ell_n(\gamma,\lambda) - \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\ell_n(\gamma,\lambda)])^2\right] \\
&\quad = \underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}\left[(\ell_n(\gamma,\lambda) - \underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}[\ell_n(\gamma,\lambda)] + \underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}[\ell_n(\gamma,\lambda)] - \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\ell_n(\gamma,\lambda)])^2\right] \\
&\quad = \underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathrm{Cov}}(\ell_n(\gamma,\lambda)) + \left(\underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}[\ell_n(\gamma,\lambda)] - \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\ell_n(\gamma,\lambda)]\right)^2.
\end{aligned}
$$

If we assume that the third–order posterior cumulant

$$
\underset{\mathbb{P}(\gamma|\mathcal{X})}{\mathbb{E}}\left[(\gamma - \underset{\mathbb{P}(\gamma|\mathcal{X})}{\mathbb{E}}[\gamma]) \left(\underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}[\ell_n(\gamma,\lambda)] - \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\ell_n(\gamma,\lambda)]\right)^2\right]
$$

is $\tilde{O}_p(N^{-2})$ — as it would be if the corresponding moments of $\mathbb{P}(\gamma|\mathcal{X})$ obeyed an expansion like Theorem 1 — then Eq. 10 follows.

F.1 Derivatives of posterior expectations

Consider an exponentially tilted distribution,

$$\mathbb{P}(\theta|t) := \frac{\exp\left(\ell(\theta) + t\phi(\theta)\right)}{\int \exp\left(\ell(\theta') + t\phi_k(\theta')\right) d\theta'}$$

which we assume is defined for all t in a neighborhood of the zero vector. When we can exchange the order of integration and differentiation,

$$\left.\frac{d \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right]}{dt}\right|_{t} = \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\phi(\theta)\right] - \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right] \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\right].$$

Applying the previous formula to itself gives

$$\begin{aligned}
\left.\frac{d^2 \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right]}{dt^2}\right|_{t} = {} & \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\phi(\theta)\phi(\theta)\right] - \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\phi(\theta)\right] \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\right] - \\
& \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right] \left(\underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\phi(\theta)\right] - \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\right] \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\right] \right) - \\
& \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\right] \left(\underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\phi(\theta)\right] - \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right] \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[\phi(\theta)\right] \right).
\end{aligned}$$

If, for some t_0, we set

$$\begin{aligned}
\bar{g}(\theta) &:= g(\theta) - \underset{\mathbb{P}(\theta|t_0)}{\mathbb{E}}\left[g(\theta)\right] \\
\bar{\phi}(\theta) &:= \phi(\theta) - \underset{\mathbb{P}(\theta|t_0)}{\mathbb{E}}\left[\bar{\phi}(\theta)\right],
\end{aligned}$$

then the above formulas simplify at t_0 to

$$\left.\frac{d \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right]}{dt}\right|_{t=t_0} = \underset{\mathbb{P}(\theta|t_0)}{\mathbb{E}}\left[\bar{g}(\theta)\bar{\phi}(\theta)\right] \quad \text{and} \quad \left.\frac{d^2 \underset{\mathbb{P}(\theta|t)}{\mathbb{E}}\left[g(\theta)\right]}{dt^2}\right|_{t=t_0} = \underset{\mathbb{P}(\theta|t_0)}{\mathbb{E}}\left[\bar{g}(\theta)\bar{\phi}(\theta)\bar{\phi}(\theta)\right].$$

Importantly, the validity of the preceding expression for the second derivative requires that the centering is at a fixed t_0 — the same t_0 at which the derivative is evaluated — and does not depend on t.

G Experiment details and additional experiments

G.0.1 Measuring sampling error

Since our experiments use a finite M, $\hat{V}^{\mathrm{Bayes}}$ and $\hat{V}^{\mathrm{IJ}}$ are MCMC estimators of V^{Bayes} and V^{IJ}, respectively. Similarly, our B bootstrap samples are thought of as a random draw from the set of $B_{max} := N^N/N!$ possible bootstrap samples, and so $\hat{V}^{\mathrm{Boot}}$ is a "doubly Monte-Carlo" estimator of $V^{\mathrm{Boot}} := \lim_{B \to B_{max}} \lim_{M \to \infty} \hat{V}^{\mathrm{Boot}}$. In general, for $B < B_{max}$, we expect bootstrap sampling variability to remain even in $\lim_{M \to \infty} \hat{V}^{\mathrm{Boot}}$. Similarly, for $M < \infty$, we expect MCMC sampling variability to remain even in $\lim_{B \to B_{max}} \hat{V}^{\mathrm{Boot}}$. We will shortly discuss how to estimate these Monte Carlo errors.

Even when the conditions of Theorem 2 are met and the IJ covariance is consistent, we do not expect $V^{\mathrm{IJ}} = V^{\mathrm{Boot}}$ even with $B = B_{max}$ and $M = \infty$, since we have a finite number N of data points, and the bootstrap and IJ covariance are formally distinct (though each consistent) estimators of V^g. The difference $V^{\mathrm{IJ}} - V^{\mathrm{Boot}}$ should decrease as N grows, but, for any finite N, the difference can be expected to exceed Monte Carlo error for sufficiently large M and B. Similarly, even in the correctly specified case, we do not expect $V^{\mathrm{Bayes}} = \hat{V}^g$, as, again, these are distinct (though consistent) estimators of $\mathcal{I}^{-1}$.

One might attempt to additionally account for frequentist sampling variability of the original data from the target population in V^{IJ} or V^{Boot} as estimators of V^g, or of V^{Bayes} as an estimator of $\mathcal{I}^{-1}$. However, we expect the quantities V^{IJ}, V^{Boot}, and V^{Bayes} to be highly correlated under random sampling of the data from the population distribution, and it is necessary to take this correlation into account when computing the variability of, say, the IJ "error" $\hat{V}^{\mathrm{IJ}} - \hat{V}^{\mathrm{Boot}}$. The authors do not see a simple, computationally tractable way to account for this correlation, so we neglect this error in our experiments, and account only for the Monte Carlo error of the preceding paragraph. Since we consider the IJ covariance to succeed when the difference $\hat{V}^{\mathrm{IJ}} - \hat{V}^{\mathrm{Boot}}$ is small relative to random variability, neglecting the frequentist sampling error is conservative.

To estimate the Monte Carlo standard errors, we model the individual entries of the matrices $\hat{V}^{\mathrm{Bayes}}$, $\hat{V}^{\mathrm{IJ}}$, and $\hat{V}^{\mathrm{Boot}}$ as normal distributions with Monte Carlo standard deviations given by the matrices Ξ^{bayes}, Ξ^{IJ}, and Ξ^{boot}, respectively.[1] To estimate Ξ^{bayes} and Ξ^{IJ}, we block bootstrap the MCMC chains using block sizes that are large relative to estimates of the MCMC mixing times. Note that block bootstrapping here requires only re-arranging the original MCMC chains, which is fast, and does not require drawing any new MCMC samples. To estimate Ξ^{boot}, we use the fact that each bootstrap sample $\mathscr{G}^b$ from Algorithm 2 contains independent MCMC errors, and so the marginal distribution of the $\mathscr{G}^b$ can be used to estimate the variability both of the bootstrap sampling and the MCMC. We do so using the delta method, as follows. For any two scalar random variables, t_1 and t_2 drawn

[1]Since both $\hat{V}^{\mathrm{Bayes}}$ and $\hat{V}^{\mathrm{IJ}}$ are based on the same MCMC samples, their Monte Carlo errors are correlated, typically positively. However, to avoid cumbersome exposition of the experimental results, we will report standard errors for $\hat{V}^{\mathrm{Bayes}}$ and $\hat{V}^{\mathrm{IJ}}$ as if they were independent, since accounting for this correlation will tend to reduce the standard errors of the difference $\hat{V}^{\mathrm{Bayes}} - \hat{V}^{\mathrm{IJ}}$, and the difference $\hat{V}^{\mathrm{Bayes}} - \hat{V}^{\mathrm{IJ}}$ tends to be large in our experiments, even relative to the conservative estimate of independent standard errors.

according to a joint distribution $p(t_1, t_2)$, we can define $h(a, b, c) := a - bc$ and write

$$\begin{aligned}\underset{p(t_1,t_2)}{\mathrm{Cov}}(t_1, t_2) &= \underset{p(t_1,t_2)}{\mathbb{E}}[t_1 t_2] - \underset{p(t_1,t_2)}{\mathbb{E}}[t_1] \underset{p(t_1,t_2)}{\mathbb{E}}[t_2] \\ &= h\left(\underset{p(t_1,t_2)}{\mathbb{E}}[t_1 t_2], \underset{p(t_1,t_2)}{\mathbb{E}}[t_1], \underset{p(t_1,t_2)}{\mathbb{E}}[t_2]\right).\end{aligned}$$

By applying the delta method to the function h, we can estimate the sampling variability of $\underset{p(t_1,t_2)}{\mathrm{Cov}}(t_1, t_2)$ from the covariance of the vector $(t_1 t_2, t_1, t_2)$, which, in turn, can be estimated from draws of t_1 and t_2. We apply the delta method in this way to each entry of the matrix $\hat{V}^{\mathrm{Boot}}$.

G.1 Applied regression and modeling textbook details

	Fixed effects only	Random and fixed effects
Linear regression	398	168
Logistic regression	192	10

Models

	Cross-covariance estimate	Variance estimate
Log scale parameter	191	62
Regression parameter	325	190

Covariances to estimate

Table 2: Models from Gelman and Hill, 2006

As discussed in Section 4.2, for the ARM experiments we needed to make decisions about what was an exchangeable unit. For models with no random effects, we define an exchangeable unit to be a tuple containing a particular response and a regressor. For random effects models, we had to choose an exchangeable unit. When the random effect had many distinct levels, we considered the groups of observations within a level to be a single exchangeable unit. For models whose random effects had few distinct levels, we took the exchangeable unit to either be a more fine-grained partition of the data, or, in some cases treated the observations as fully independent.

Although the posterior covariance is not of primary interest, we compute it to investigate the possibility of differences between the frequentist and posterior covariance estimates. Given that most of the models in the ARM textbook are carefully selected by highly experienced statisticians, one might expect there to be few instances of misspecification. Since the models are additionally mostly low-dimensional, we might expect for the posterior covariance to typically provide a good approximation to the frequentist covariance by the Bernstein von-Mises theorem.

To investigate this intuition, in Figure 1 we reproduce Figure 4 but comparing the bootdstrap and Bayesian posterior covariance. The metric plotted is the posterior version of $\Delta_{ij}^{\mathrm{IJ}}$ in Eq. 15:

$$\Delta_{ij}^{\mathrm{Bayes}} := \frac{\hat{V}_{ij}^{\mathrm{Bayes}} - \hat{V}_{ij}^{\mathrm{Boot}}}{\left|\hat{V}_{ij}^{\mathrm{Boot}}\right| + \Xi_{ij}^{\mathrm{boot}}}.$$

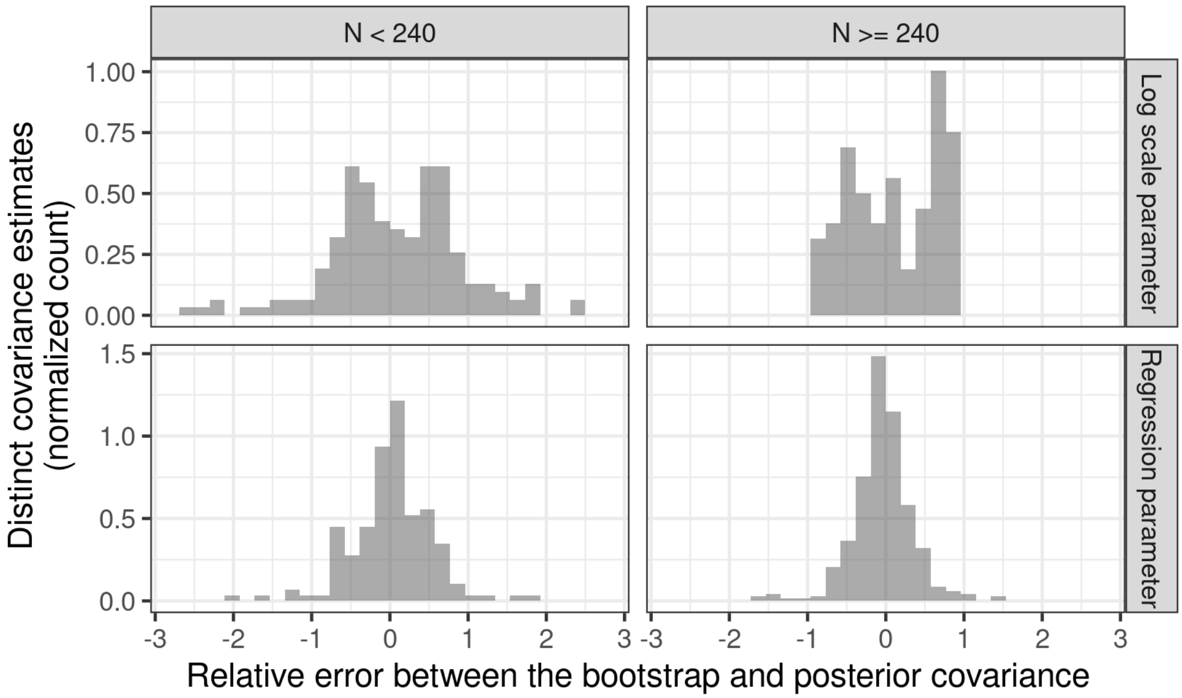

Figure 1: The distribution of the relative errors Δ^{Bayes}. The "Log scale parameter" panel includes all variances or covariances that involve at least one log scale parameter.

Comparing with Figure 4, we see that the above intuition is mostly correct, and that the frequentist and posterior covariances are largely similar. The notable exception is for log scale parameters in models with relatively large datasets, for which the IJ provides a markedly better approximation to the bootstrap than does the posterior covariance. This suggests the possibility of using the difference between IJ and posterior covariances as a diagnostic for model misspecification, as is sometimes done in the frequentist literature (White, 1982). We leave this possibility as an interesting avenue for future work.

G.2 Multilevel regression and poststratification

For a more complex model, we consider an application of "multilevel regression and post–stratification," or MrP (Park, Gelman, and Bafumi, 2004). We studied an example from Lopez-Martin, Phillips, and Gelman (2022) in which a subsample of a national survey is used to infer a prediction on the entire population. Since the textbook example involves simulated subsampling, a ground truth is available for comparison.

MrP models can be motivated by the following survey sampling problem. Let z_* denote a vector of regressors, and y_* a binary survey response. Suppose we observe N_S observations of the form $x_n = (z_n, y_n)$ from a "survey population," and N_T observations consisting only of z_m from a "target population." We assume that $\underset{\mathbb{P}(y_*|z_*)}{\mathbb{E}}[y_*]$ is the same in both the survey and target populations, but that the marginal distributions of z_* may differ, perhaps due to biased survey sampling. Let $\mathbb{P}_T(\cdot)$ denote the target population distribution. We would like to use the observed data to estimate $\underset{\mathbb{P}_T(y_*)}{\mathbb{E}}[y_*] = \underset{\mathbb{P}_T(z_*)}{\mathbb{E}}\left[\underset{\mathbb{P}(y_*|z_*)}{\mathbb{E}}[y_*]\right]$ in the target population. The approach of MrP is to use the survey data to form a regression estimate of $z_* \mapsto \underset{\mathbb{P}(y_*|z_*)}{\mathbb{E}}[y_*]$ (the "multilevel regression" part of MrP) and then to average the regression's predictions using the target population regressors (the "post–stratification" part of MrP).

A typical MrP model assumes that the expectation of the response conditional on the regressors is given a finitely parameterized regression model. For example, one might specify a logistic regression model for a binary survey response. The regression model gives an explicit expression for $\mathbb{E}[y|z,\theta]$ which, by assumption, holds in both the survey and target populations. Typically, the MrP model also specifies hierarchical, or "multilevel" prior structure on the parameters, possibly including unknown hyperparameters. For compactness of notation we assume that these unknown hyperparameters are also included in θ, even if they are not used directly in the computation of $\mathbb{E}[y|z,\theta]$. For any particular value of θ we can form the "poststratified" estimator of $\underset{\mathbb{P}_T(y_*)}{\mathbb{E}}[y_*]$, which is our quantity of interest:

$$g(\theta) = \frac{1}{N_T}\sum_{m=1}^{N_T} \mathbb{E}[y_*|z_m,\theta]. \tag{25}$$

The expression for $g(\theta)$ is motivated by the observation that if $\mathbb{E}[y_*|z_m,\theta] = \mathbb{E}[y_*|z_m]$ at our particular θ, then $g(\theta) \approx \underset{T(y_*)}{\mathbb{E}}[y_*]$ by the law of large numbers, since z_m are drawn marginally from the target distribution of regressors. The Bayesian estimator integrates out our uncertainty in the θ, taking $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ as our posterior estimate of the unknown $\underset{T(y_*)}{\mathbb{E}}[y_*]$. See Park, Gelman, and Bafumi (2004) and Lopez-Martin, Phillips, and Gelman (2022) for more details.

To assess our ability to estimate V^g in MrP models, we replicate a semi–simulated analysis from Chapter 1 of the MrP tutorial textbook, Lopez-Martin, Phillips, and Gelman (2022). Although this MrP analysis is based on real data, the survey population consists of a sample taken uniformly at random from a *superpopulation* for which complete data — both regressors and responses — is available. The MrP estimator $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$, which is formed using response variables only from the sampled survey population, can then be compared to the known ground truth: namely, the average response in the superpopulation. Lopez-Martin, Phillips, and Gelman (2022) use the superpopulation to assess the accuracy of MrP methods. In our case, we use the superpopulation to estimate the ground truth V^g by measuring the sampling variability of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ under repeated independent draws of the survey population.

The superpopulation dataset provided by Lopez-Martin, Phillips, and Gelman (2022) consists of 59,810 pairs of survey responses and regressors, from which we sample 5,000 uniformly at random without replacement to form simulated survey populations. The response y_* is an indicator of whether a respondent supports coverage of abortions in insurance plans, taken from the 2018 Cooperative Congressional Election Study (Schaffner, Ansolabehere, and Luks, 2018). The provided regressors, z_*, consist of discretized demographic information measuring age, gender, ethnicity, education, and state. We use the model given by Chapter 1 of Lopez-Martin, Phillips, and Gelman (2022), which consisted of a hierarchical logistic regression with unknown hyperparameter variances. We let our parameter θ include both logistic regression coefficients and hyperprior variances; in total, the dimension of the parameter space was $D_\theta = 129$.

The details of our sampling and estimation procedure were as follows. A draw of the survey data $\mathcal{X}$ is produced by selecting 5,000 regressor / response pairs $x_n = (z_n, y_n)$ uniformly at random without replacement from the 59,810 observations in the superpopulation. For each such $\mathcal{X}$, we can compute $\mathbb{P}(\theta|\mathcal{X})$ and so $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$, using the 59,810 − 5,000 remaining superpopulation observations as the target population in $g(\theta)$.[2] By simulating multiple draws of $\mathcal{X}$ and computing $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ for each, we can estimate the ground truth V^g. In this way, we simulated 101 different survey datasets $\mathcal{X}$. For each dataset, we drew $M = 18{,}000$ MCMC samples from each posterior using the `R` package `brms` (Bürkner, 2018). We then used these samples to compute an MCMC estimate of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ on each, the variance of which across distinct simulated values of $\mathcal{X}$ provides a ground–truth estimate of V^g. On each set of MCMC draws, we computed $\hat{V}^{\mathrm{IJ}}$ and $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Var}}(g(\theta))$. We randomly chose one of the samples as a basis for 101 bootstrap replicates to compute a bootstrap variance estimate $\hat{V}^{\mathrm{Boot}}$.

As shown in Figure 2, the IJ covariance and bootstrap covariance are nearly identical. In this case, the Bayesian posterior variance also coincides with the frequentist variance. The total computation time for all bootstrap samples was 12 minutes, and the computation for the IJ covariance on the same set of MCMC draws used for the bootstraps was 15 seconds.[3]

As we show in Appendix G.2.1, on this problem the MAP estimate provides a poor estimate of the posterior expectation, likely due to the relatively high dimensionality of θ and the nonlinearity of the model, particularly through estimation of the random effect variances. Nevertheless, since $N/D_\theta \approx 38.76$, it is plausible that the posterior is concentrated enough in the nuisance parameters to avoid the challenges for IJ in the high–dimensional regime we discussed in Section 3.3. In Appendix G.2.1, we also compare with marginal maximum likelihood estimates provided by the `R` function `lmer` (Bates et al., 2015), noting that `lmer` does not naturally provide estimates of V^g for a complicated quantity depending jointly on the random and fixed effects.

G.2.1 Multilevel regression and poststratification details

[2]Technically there will be additional variability due to the random selection of the target regressors in Eq. 25, but we are justified in neglecting this variability since the superpopulation is so large.

[3]The computation time for the IJ covariance was dominated by a single call to `brms::posterior_linpred`, which took 10 of the 15 seconds. The function `brms::posterior_linpred` is only computing the log odds for each observation and each MCMC draw, but is doing so with a sequence of nested loops rather than vectorized operations, due to the idiosyncrasies of `rstan` data objects. In a different MCMC environment, this computation could presumably be done much more efficiently.

Figure 2: Covariance estimates for the MrP model. The dots are the estimates computed on the single dataset chosen as the basis for bootstrapping. The boxplots show the spread of the IJ and posterior covariance across simulations, which includes both frequentist and MCMC variability. The horizontal line shows the true sampling variance as estimated by simulation.

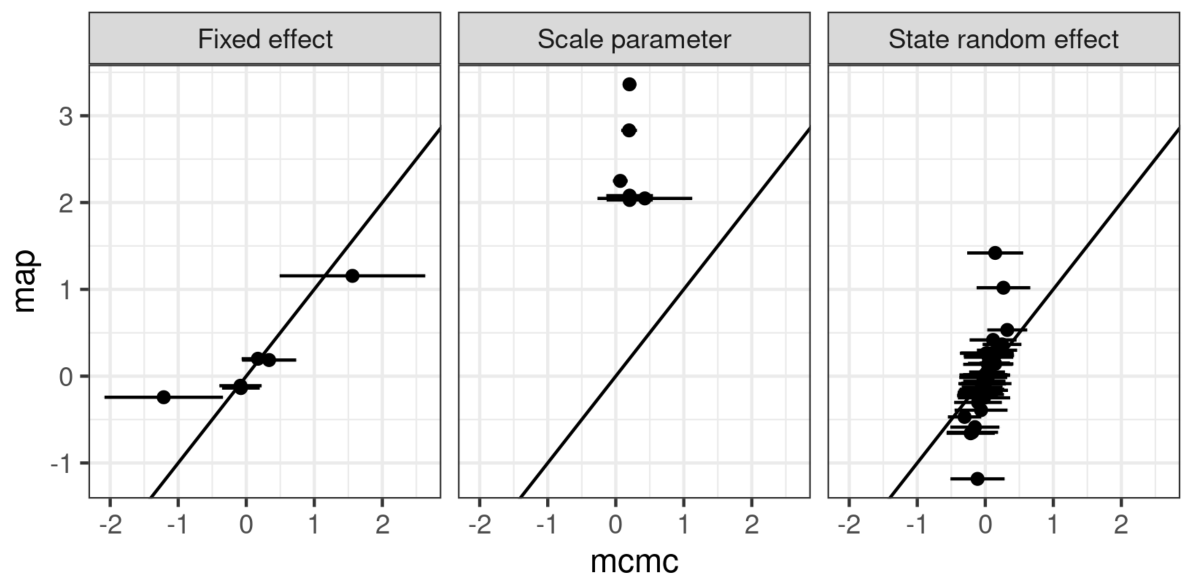

Figure 3: MAP versus MCMC for the MrP model

In Figure 3 we see that the MAP estimator provides poor point estimates in the MrP model, likely due in part to the large number of random effects to fit. We note that the `stan::optimizing` function in `R` performed poorly, and we had find the MAP by hand using a careful initialization and small step size in order to avoid divergent estimates.

Although it is not strictly speaking a MAP estimate, the natural optimization–based

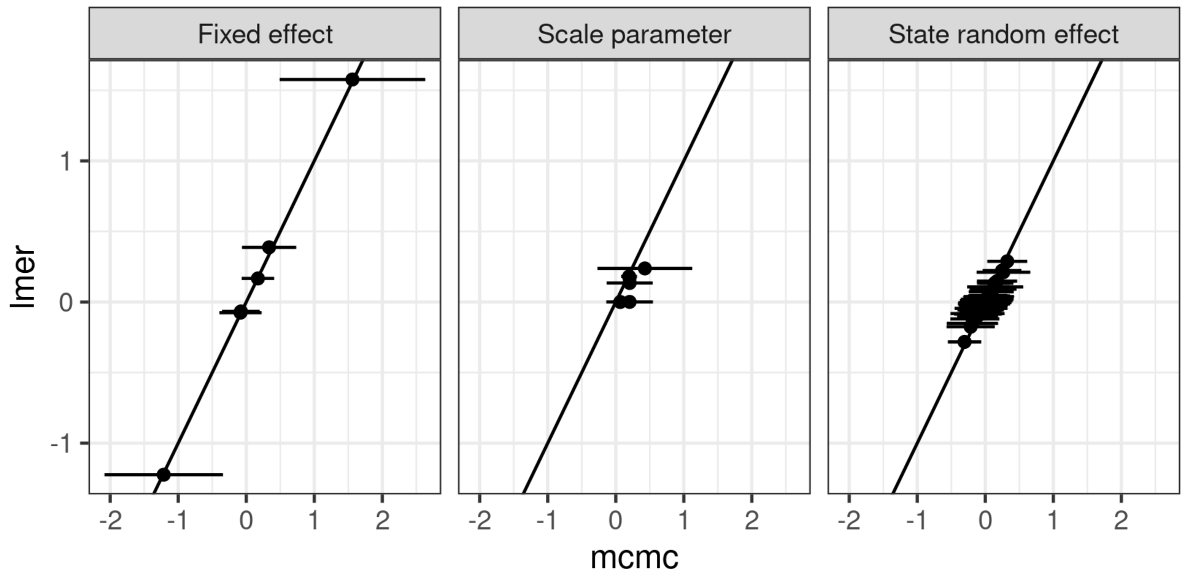

Figure 4: LMER versus MCMC for the MrP model

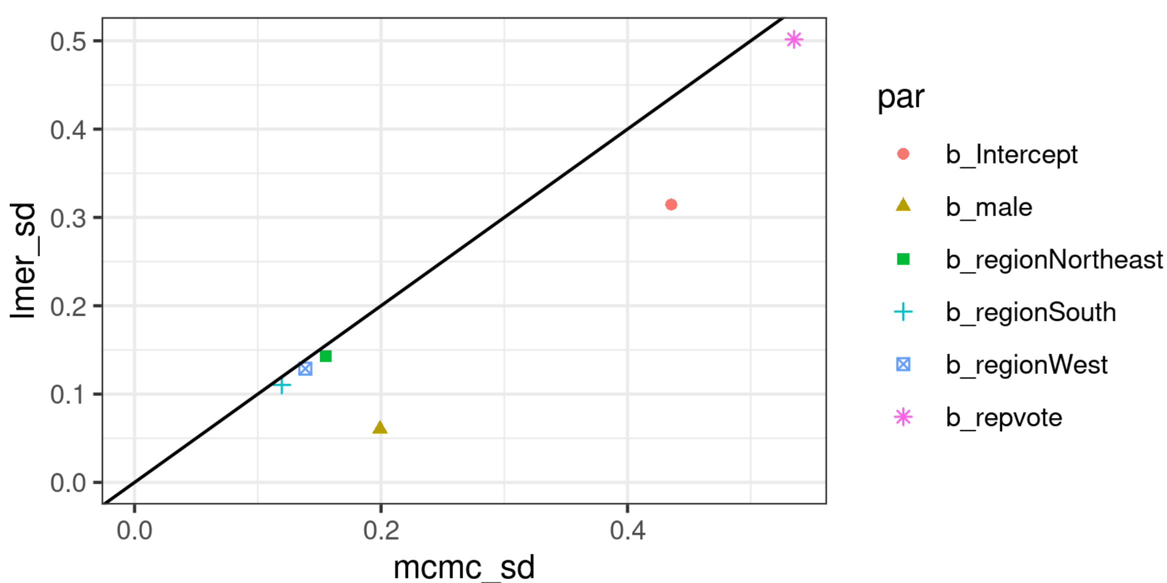

Figure 5: LMER versus MCMC standard errors for the MrP model

competitor for MCMC is not the MAP estimator but marginal maximum likelihood estimation, such as the `lmer` function in `R`. As shown in Figure 4, `lmer` provides better point estimates than MAP estimation on the MrP model. However, as shown in Figure 5, the standard error estimates are provided by `lmer` can differ considerably from the Bayesian posterior estimates, and most parameters do not have standard errors reported automatically. In principle, one may be able combine the frequentist sampling estimates of each parameter to estimate the joint sampling distribution between the fixed in order to estimate the sampling variance of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$, but doing so would be quite cumbersome given the information exposed by `lmer`. In contrast, the MCMC and IJ covariance estimates are extremely straightforward to compute.

H Additional closed-form examples

In this section we consider some simple illustrative examples. All the models in this section will be toy problems for which the IJ is not necessary, but they will serve to build intuition for harder, real-life problems.

In each of these examples we will go through the tedious effort to explicitly compute $\psi(x_n)$ and the limit of its sample variance. We stress, however, that we are only doing so for illustrative purposes. In practice, one would simply output $\ell(x_n|\theta)$ from the MCMC algorithm and estimate the posterior covariances using MCMC draws, for which none of the below analytic derivations would be necessary.

H.0.1 Sample mean

Suppose we observe scalar data, $x_n \in \mathbb{R}$, for $n = 1, \dots, N$. We assume that x_n and x_n^2 obey laws of large numbers, and that x_n itself obeys a central limit theorem, with

$$\begin{aligned}
\hat{\mu} &:= \frac{1}{N}\sum_{n=1}^{N} x_n \xrightarrow[N\to\infty]{prob} \mu \\
\hat{V}_x &:= \frac{1}{N}\sum_{n=1}^{N} (x_n - \hat{\mu})^2 \xrightarrow[N\to\infty]{prob} V_x \\
&\sqrt{N}(\hat{\mu} - \mu) \xrightarrow[N\to\infty]{dist} \mathcal{N}\left(\cdot|0, V_x\right).
\end{aligned}$$

Suppose we use the data to compute a Bayesian posterior using the following model, with fixed prior parameter σ_θ, and a fixed variance σ^2:

$$\begin{aligned}
\theta &\sim \mathcal{N}(\cdot|0, \sigma_\theta^2) \\
x_n|\theta &\overset{iid}{\sim} \mathcal{N}(\cdot|\theta, \sigma^2).
\end{aligned}$$

We will take our parameter of interest to be θ itself, so $g(\theta) = \theta$. The posterior has a closed form, given by

$$\begin{aligned}
\hat{\theta} &:= \frac{\sigma^{-2}\sum_{n=1}^{N} x_n}{N\sigma^{-2} + \sigma_\theta^{-2}} = \frac{N}{N + \sigma^2/\sigma_\theta^2}\hat{\mu} \\
p(\theta|\mathcal{X}) &= \mathcal{N}\left(\cdot|\hat{\theta}, \frac{\sigma^2}{N + \sigma^2/\sigma_\theta^2}\right).
\end{aligned}$$

Consequently,

$$\begin{aligned}
N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Var}}\left(\theta\right) &\xrightarrow[N\to\infty]{prob} \sigma^2 \\
\sqrt{N}\left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\theta\right] - \mu\right) &\xrightarrow[N\to\infty]{dist} \mathcal{N}\left(\cdot|0, V_x\right).
\end{aligned}$$

In this case, if $V_x \neq \sigma^2$—that is, if the model is misspecified— then the limiting variance of the frequentist and Bayesian distributions are different.

How does the IJ perform in this context? In this case, the log likelihood is given up to constants in θ by

$$\ell(x_n|\theta) = -\frac{1}{2}\sigma^{-2}\left(x_n^2 - 2x_n\theta + \theta^2\right) + Constant.$$

So the influence scores are given by

$$\begin{aligned}
\psi_n = & -N\frac{1}{2}\sigma^{-2}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta, -2x_n\theta + \theta^2\right) \\
& = N\sigma^{-2}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta\right)x_n - N\frac{1}{2}\sigma^{-2}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta, \theta^2\right).
\end{aligned}$$

To evaluate the second covariance, use the fact that, by normality,

$$\begin{aligned}
&\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[(\theta - \hat{\theta})^3\right] = 0 \qquad \text{and} \qquad \theta^2 = (\theta - \hat{\theta})^2 + 2\hat{\theta}\theta - \hat{\theta}^2 \Rightarrow \\
&\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta, \theta^2\right) = 2\hat{\theta}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta\right).
\end{aligned}$$

Combining the results, we see that

$$\begin{aligned}
\psi_n &= N\sigma^{-2}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta\right)(x_n - \hat{\theta}) \\
&= \frac{N}{N + \sigma^2/\sigma_\theta^2}\left(x_n - \frac{N}{N + \sigma^2/\sigma_\theta^2}\hat{\mu}\right).
\end{aligned}$$

Since $N/(N + \sigma^2/\sigma_\theta^2) \to 1$, the sample variance of ψ_n converges to V_x as desired, despite the fact that the underlying model was misspecified.

H.0.2 Overdispersed Poisson

Let us now consider another simple non-normal misspecified model, an over-dispersed Poisson distribution.

$$\begin{aligned}
\text{Model:} \qquad & \theta \sim \mathrm{Gamma}(a, b) \\
& x_n \overset{iid}{\sim} \mathrm{Poisson}(\theta) \\
\text{Data generating process (reality):} \qquad & \theta = \theta_\infty \\
& \underset{\mathbb{F}(\mathcal{X})}{\mathbb{E}}\left[x_n\right] = \theta_\infty \\
& \underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}}\left(x_n\right) = \sigma^2 > \theta_\infty.
\end{aligned}$$

The posterior has a closed form:

$$
\begin{aligned}
\bar{x} &:= \frac{1}{N}\sum_{n=1}^{N} x_n \\
\hat{\sigma}^2 &:= \frac{1}{N}\sum_{n=1}^{N} x_n^2 - \bar{x}^2 \\
\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\theta\right] &= \frac{N\bar{x}+a}{N+b} \\
\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Var}}\left(\theta\right) &= \frac{N\bar{x}+a}{(N+b)^2} \Rightarrow \\
\underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}}\left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\theta\right]\right) &= \frac{N\sigma^2}{(N+b)^2}.
\end{aligned}
$$

Because x_n is over-dispersed, the posterior variance does not match the frequentist variance $\underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}}\left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\theta\right]\right)$, even asymptotically.

$$
\begin{aligned}
N\underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}}\left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\theta\right]\right) &\xrightarrow[N\to\infty]{prob} \sigma^2 \\
N\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Var}}\left(\theta\right) &\xrightarrow[N\to\infty]{prob} \theta_\infty < \sigma^2.
\end{aligned}
$$

The model and MAP are given by

$$
\begin{aligned}
\hat{\mathscr{L}}(x|\theta) &= \frac{1}{N}\sum_{n=1}^{N}\left(x_n \log\theta - \theta\right) + C \\
\log\pi(\theta) &= (a-1)\log\theta - b\theta \Rightarrow \\
\hat{\theta} &= \frac{N\bar{x}+a-1}{N+b} \\
\ell_{(1)}(x|\theta) &= \frac{x_n}{\theta} - 1 \\
\hat{\mathscr{L}}_{(2)}(x|\theta) &= -\frac{\bar{x}}{\theta^2}.
\end{aligned}
$$

Here, the MAP and posterior expectation differ by a factor of $1/(N+b)$, though, as expected, the difference is $o_p(N^{-1/2})$. We can now compute the sandwich covariance estimator:

$$
\begin{aligned}
\hat{\Sigma} &= \frac{\hat{\sigma}^2}{\hat{\theta}^2} \\
\hat{\mathcal{I}} &= -\hat{\mathscr{L}}_{(2)}(x|\hat{\theta}) = \frac{\bar{x}}{\hat{\theta}^2} \\
\hat{V}^g &= \hat{\mathcal{I}}^{-1}\hat{\Sigma}\hat{\mathcal{I}}^{-1} = \frac{\hat{\theta}^2}{\bar{x}^2}\hat{\sigma}^2 = \left(\frac{N\bar{x}+a-1}{(N+b)\bar{x}}\right)^2\hat{\sigma}^2.
\end{aligned}
$$

As expected, the limits of $N \underset{\mathbb{F}(\mathcal{X})}{\mathrm{Var}} \left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}} [\theta] \right)$ and $\hat{V}^g$ are the same.

The influence function is given by

$$
\begin{aligned}
\psi(x_n) &= N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\theta, x_n \log \theta - \theta \right) \\
&= N x_n \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\theta, \log \theta \right) - N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Var}} \left(\theta \right) \\
&= \frac{N}{N+b} x_n + C.
\end{aligned}
$$

where C is the same for every x_n, and where we have used the posterior covariance $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\theta, \log \theta \right)$ for the Gamma distribution, which can be found by taking the mixed second-order partial derivative of the log partition function. It follows that

$$
V^{\mathrm{IJ}} = \frac{1}{N} \sum_{n=1}^{N} \psi(x_n)^2 - \bar{\psi}^2 = \frac{N^2}{(N+b)^2} \hat{\sigma}^2,
$$

which is also a consistent estimator of $\sigma^2 = V^g$.

H.0.3 Multivariate regression

We now consider multivariate regression with unknown variance. This will provide a simple multivariate example, draw a connection with robust frequentist estimators, and allow us to discuss a point about conditioning on part of the exchangeable unit.

The model and data generating process are

$$
\begin{aligned}
\text{Model:} \qquad & \sigma^{-2} \sim \mathrm{Gamma}(a, b) \\
& \beta \sim \mathcal{N}(\cdot | 0, \sigma_\beta^2 I_D) \\
& y_n \overset{iid}{\sim} \mathcal{N}(\cdot | \beta^T z_n, \sigma^2) \\
\text{Data generating process (reality):} \qquad & \text{The moments of } y_n z_n,\ z_n z_n^T, \text{ and } y_n^2 z_n z_n^T \\
& \text{are finite and obey a LLN, with} \\
& \text{invertible } \underset{\mathbb{F}(\mathcal{X})}{\mathbb{E}} \left[z_n z_n^T \right].
\end{aligned}
$$

Here, $\theta = (\beta, \sigma)$ and $x_n = (y_n, z_n)$. We will consider $g(\theta) = \beta$. The full posterior has a closed form, but we will only require the conditional posterior $p(\beta|x,\sigma)$.

$$\hat{S}_z := \frac{1}{N}\sum_{n=1}^{N} z_n z_n^T + \frac{\sigma_\beta^2}{N} I_D$$

$$\hat{\beta} := \hat{S}_z^{-1}\left(\frac{1}{N}\sum_{n=1}^{N} z_n y_n\right)$$

$$p(\beta|x,\sigma) = \mathcal{N}\left(\beta|\hat{\beta}, \sigma^2 \hat{S}_z\right)$$

$$\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[\beta] = \hat{\beta}$$

$$\mathop{\mathrm{Cov}}_{\mathbb{P}(\theta|\mathcal{X})}(\beta) = \frac{\sigma^2}{N}\hat{S}_z^{-1}.$$

In order to consider the frequentist variance of $\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[\beta]$, we have to specify what is random. Since $x_n = (z_n, y_n)$, the analysis of this paper appears to require that both z_n and y_n are random. However, the generative process for the regressors z_n are not part of the Bayesian model given above. Indeed, the generating process for the regressors is hardly ever specified in Bayesian regression because the data generating process for z_n is ancillary to the parameters of interest, θ. However, precisely because $p(z_n)$ is ancillary for θ, one could resolve this apparent inconsistency by imagining including any data generating process $p(z_n)$ to the Bayesian model, and observing that the Bayesian posterior is unchanged no matter what $p(z_n)$ is.

Asymptotically, the difference between random z_n and fixed z_n does not matter since sample moments over z_n converge to their population counterparts. However, in finite sample, the difference can be important, and it is easier to bootstrap pairs (z_n, y_n) than keep z_n fixed and resample only y_n. (We will not consider methods such as the residual bootstrap here.) Unless otherwise stated, we will thus take the view in the present paper that the full exchangeable pair is random.

Because of the term $\hat{S}_z^{-1}$ and the randomness of z_n, the variance $N\mathop{\mathrm{Var}}_{\mathbb{F}(\mathcal{X})}\left(\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[\beta]\right)$ does not have a closed form. However, by a CLT applied to $\sqrt{N}\frac{1}{N}\sum_{n=1}^{N} z_n(y_n - \hat{\beta}^T z_n)$ and Slutsky's theorem, the variance is consistently estimated by

$$\hat{\epsilon}_n := y_n - \hat{\beta}^T z_n$$

$$N\mathop{\mathrm{Var}}_{\mathbb{F}(\mathcal{X})}\left(\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[\beta]\right) \approx \hat{S}_z^{-1}\left(\frac{1}{N}\sum_{n=1}^{N}\hat{\epsilon}_n^2 z_n z_n^T\right)\hat{S}_z^{-1}. \tag{26}$$

If the model is correctly specified, and $\sigma = \sigma_0$, then

$$\frac{1}{N}\sum_{n=1}^{N} z_n z_n^T \hat{\epsilon}_n^2 \xrightarrow[N\to\infty]{prob} \sigma_0^2 \mathop{\mathbb{E}}_{\mathbb{F}(\mathcal{X})}\left[z_n z_n^T\right],$$

and $N\mathop{\mathrm{Var}}_{\mathbb{F}(\mathcal{X})}\left(\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[\beta]\right)$ coincides asymptotically with $N\mathop{\mathrm{Cov}}_{\mathbb{P}(\theta|\mathcal{X})}(\beta)$. However, if the residuals

$\hat{\epsilon}_n$ or their variances are correlated with z_n, then the frequentist and Bayesian covariance will not match, even asymptotically. The covariance estimate in Eq. 26 is known in the frequentist regression literature as a "robust" covariance estimate, and it is sometimes recommended when there is concern about model misspecification (e.g. Davidson and MacKinnon, 2004, Section 5.5). In the present context, it is simply the regression version of the sandwich covariance matrix V^g.

The influence function in this case is given by

$$\begin{aligned}
\psi(x_n) &= N \operatorname*{Cov}_{\mathbb{P}(\theta|\mathcal{X})}\left(\beta, -\frac{1}{2\sigma^2}\left(y_n^2 - 2\beta^T z_n y_n + \beta^T z_n z_n^T \beta\right)\right) \\
&= N \operatorname*{Cov}_{\mathbb{P}(\theta|\mathcal{X})}\left(\beta, \sigma^{-2}\beta\right) z_n y_n - \frac{N}{2} \operatorname*{Cov}_{\mathbb{P}(\theta|\mathcal{X})}\left(\beta, \sigma^{-2}\right) y_n^2 + \frac{N}{2} \operatorname*{Cov}_{\mathbb{P}(\theta|\mathcal{X})}\left(\beta, \sigma^{-2}\beta^T z_n z_n^T \beta\right).
\end{aligned}$$

First,

$$\begin{aligned}
\operatorname*{Cov}_{\mathbb{P}(\theta|\mathcal{X})}\left(\beta, \sigma^{-2}\beta\right) &= \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\left(\beta - \hat{\beta}\right)\left(\sigma^{-2}\beta - \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\sigma^{-2}\beta\right]\right)\right] \\
&= \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\left(\beta - \hat{\beta}\right)\sigma^{-2}\beta\right] \\
&= \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\sigma^{-2} \mathop{\mathbb{E}}_{p(\beta|x,\sigma)}\left[\left(\beta - \hat{\beta}\right)\beta\right]\right] \\
&= \frac{1}{N} \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\sigma^{-2}\sigma^2 \hat{S}_z^{-1}\right] \\
&= \frac{1}{N}\hat{S}_z^{-1}.
\end{aligned}$$

Now, the posterior mean $\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[\beta] = \hat{\beta}$ does not depend on σ, so

$$\begin{aligned}
\operatorname*{Cov}_{\mathbb{P}(\theta|\mathcal{X})}\left(\beta, \sigma^{-2}\right) &= \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\left(\beta - \hat{\beta}\right)\left(\sigma^{-2} - \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\sigma^{-2}\right]\right)\right] \\
&= \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\mathop{\mathbb{E}}_{p(\beta|x,\sigma)}\left[\beta - \hat{\beta}\right]\left(\sigma^{-2} - \mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\sigma^{-2}\right]\right)\right] \\
&= 0.
\end{aligned}$$

Similarly, the third-order array of third-order covariances is identically zero:

$$\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}\left[\sigma^{-2}(\beta - \hat{\beta})(\beta - \hat{\beta})(\beta - \hat{\beta})\right] = 0.$$

To use this in $\psi(x_n)$, observe that

$$\beta^T z_n z_n^T \beta = \left(\beta - \hat{\beta}\right)^T z_n z_n^T \left(\beta - \hat{\beta}\right) + 2\hat{\beta}^T z_n z_n^T \beta - \hat{\beta}^T z_n z_n^T \hat{\beta},$$

so that

$$\begin{aligned}
\frac{N}{2} \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\beta, \sigma^{-2} \beta^T z_n z_n^T \beta\right) &= N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\beta, \sigma^{-2} \hat{\beta}^T z_n z_n^T \beta\right) \\
&= N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(\beta, \sigma^{-2} \beta\right) z_n z_n^T \hat{\beta} \\
&= \hat{S}_z^{-1} z_n z_n^T \hat{\beta}.
\end{aligned}$$

Combining,

$$\psi(x_n) = \hat{S}_z^{-1} z_n \left(y_n - z_n^T \hat{\beta}\right).$$

We thus see that

$$\begin{aligned}
\bar{\iota} &= \hat{S}_z^{-1} \frac{1}{N} \sum_{n=1}^{N} z_n y_n = 0 \\
V^{\mathrm{IJ}} &= \frac{1}{N} \sum_{n=1}^{N} \left(\iota(x_n)^2 - \bar{\iota}^2\right) = \hat{S}_z^{-1} \left(\frac{1}{N} \sum_{n=1}^{N} \hat{\epsilon}_n^2 z_n z_n^T\right) \hat{S}_z^{-1},
\end{aligned}$$

which matches the frequentist covariance estimate given in Eq. 26.

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